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Aura Artifacts

E-Commerce Platform for Handcrafted Gemstone Jewelry

Final Project Report

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Web Development Project
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1. Introduction

Aura Artifacts is a premium e-commerce platform specializing in handcrafted gemstone jewelry. The platform provides an elegant and immersive online shopping experience for customers seeking unique, artisan-crafted pieces including bracelets, earrings, necklaces, rings, anklets, pendants, and sunglasses — all featuring genuine gemstones with spiritual and healing properties.

The platform is built using modern web technologies including PHP for server-side logic, MySQL for database management, HTML5 and CSS3 for responsive front-end design, and JavaScript for interactive features. The system implements a comprehensive set of features including user authentication with remember-me functionality, a full shopping cart and checkout system with credit card payment processing, product browsing with search and filtering, wishlists, order history tracking, a custom bracelet builder, and a complete administrative dashboard for managing all aspects of the store.

Key features of the Aura Artifacts platform include:

- A responsive, mobile-friendly design with dark mode support
- Secure user registration and login with password hashing (bcrypt)
- Product catalog with multiple images, categories, and detailed descriptions
- Custom bracelet builder with gemstone and charm selection
- Shopping cart with quantity management and real-time price updates
- Secure checkout with credit card payment validation
- User profile management with order history
- Wishlist functionality for saving favorite products
- Contact form with message management
- Newsletter subscription system
- Comprehensive admin dashboard with sales analytics
- Product, order, customer, and message management for administrators
- Accessibility features and size guide for enhanced user experience
- Gemstone meanings and educational content
- Google Maps integration on the contact page
- Help/FAQ section with searchable content

2. Database Design

This section highlights the database design for the Aura Artifacts platform, emphasizing the relationships between tables and the data objects they contain. The database is built on MySQL with the InnoDB storage engine, using utf8mb4 character set for full Unicode support. Foreign key constraints ensure referential integrity across all tables.

2.1 Tables

The following figures illustrate the tables utilized in MySQL for the Aura Artifacts database.

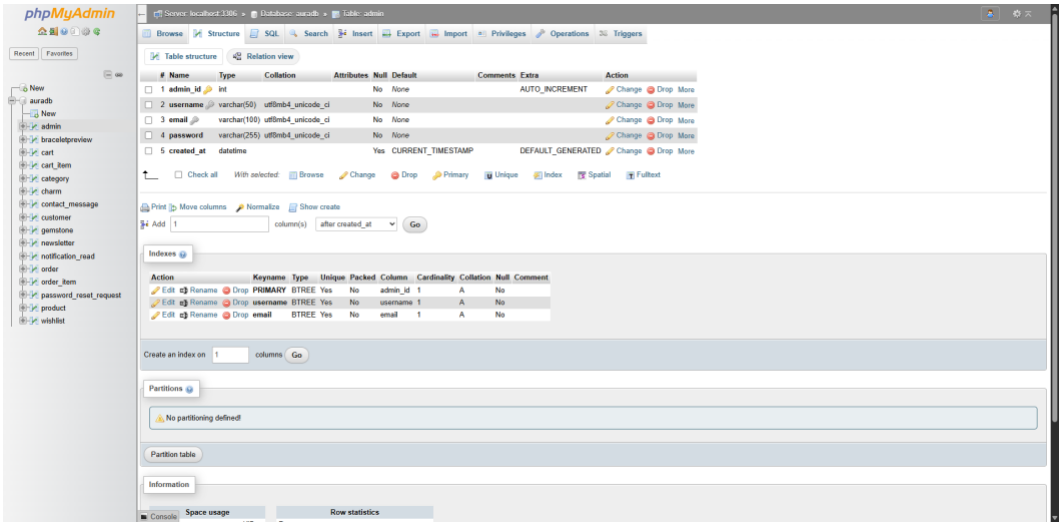


Figure 1: Admin table structure

SQL Code:

```
INSERT INTO `admin` (`admin_id`, `username`, `email`, `password`, `created_at`) VALUES
(1, 'admin', 'admin@auraartifacts.com',
 '$2y$10$92IXUNpkjO0rOQ5byMi.Ye4oKoEa3Ro9llC/.og/at2.uheWG/igi',
 CURRENT_TIMESTAMP);
```

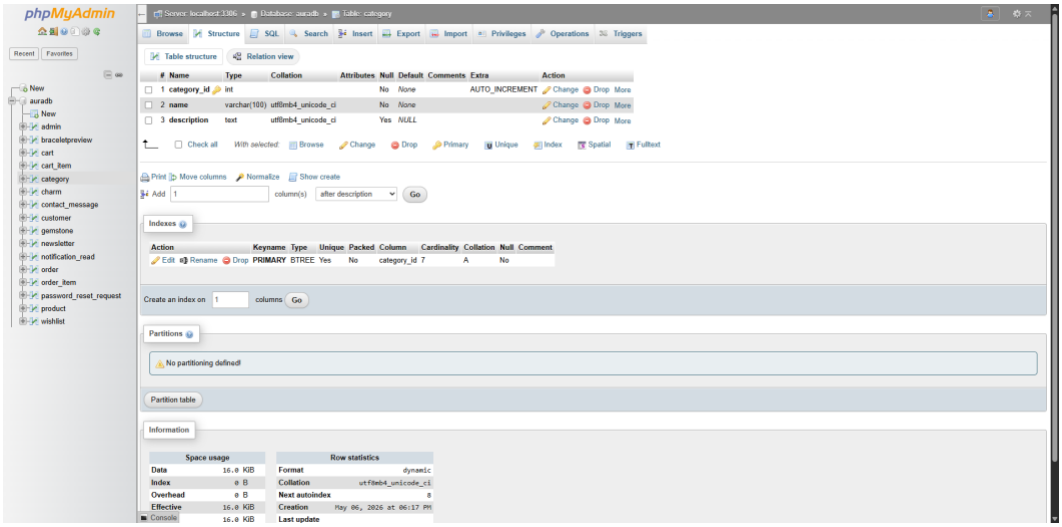


Figure 2: Category table structure

SQL Code:

```
INSERT INTO `category` (`category_id`, `name`, `description`) VALUES
(1, 'Bracelets', 'Handcrafted gemstone bracelets'),
(2, 'Earrings', 'Fine gemstone earrings'),
(3, 'Necklaces', 'Elegant necklaces and pendants'),
(4, 'Rings', 'Gemstone rings'),
(5, 'Anklets', 'Delicate anklets'),
(6, 'Pendants', NULL),
(7, 'Sunglasses', NULL);
```

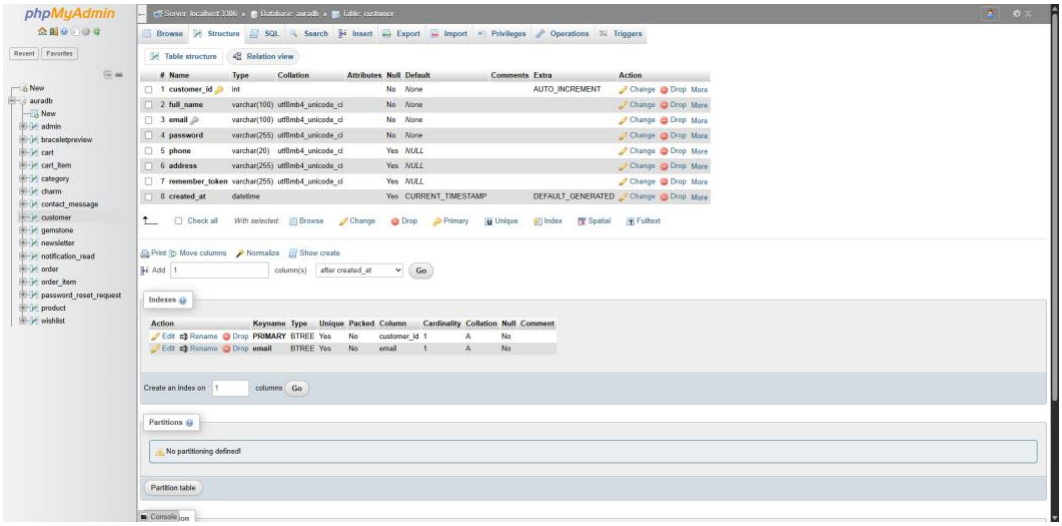


Figure 3: Customer table structure

SQL Code:

```
-- Customer table is empty on fresh install.
-- Users register through the website registration form.
-- Fields: customer_id, full_name, email, password (hashed),
--         phone, address, remember_token, created_at
```

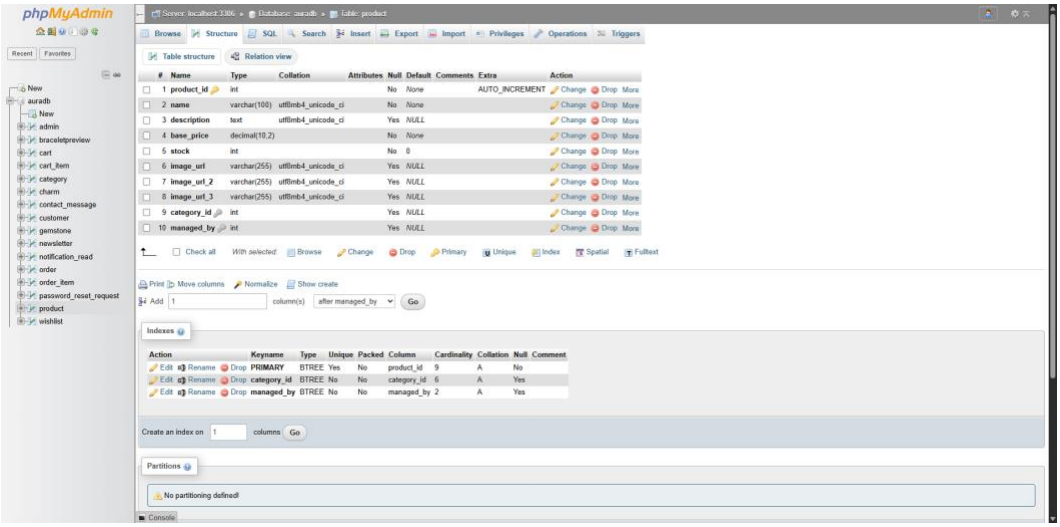


Figure 4: Product table structure

SQL Code:

```
INSERT INTO `product` (`product_id`, `name`, `description`,
  `base_price`, `stock`, `image_url`, `image_url_2`, `image_url_3`,
  `category_id`, `managed_by`) VALUES
(1, 'Pink Tourmaline Drop Earrings',
 'Genuine pear-cut Pink Tourmaline drops with brilliant-cut diamonds.
  Art Deco-inspired in rose gold.',
 450.00, 10, 'assets/images/firstpick-1.png',
 'assets/images/firstpick-2.png', 'assets/images/firstpick-3.png', 2, 1),
(2, 'Golden Radiance',
 'Premium Topaz stone set in 18K gold band.
  A stunning statement ring with timeless appeal.',
 550.00, 14, 'assets/images/secoundpick_1.png',
 'assets/images/secoundpick_2.png', 'assets/images/secoundpick_3.png', 4, 1),
```

```
(3, 'Royal Purple',
 'Stunning Amethyst stones on delicate chain.
  A vibrant pendant celebrating spiritual wisdom.',
 420.00, 5, 'assets/images/thirdpick_1.png',
 'assets/images/thirdpick_2.png', 'assets/images/thirdpick_3.png', 6, 1),
(4, 'Sapphire Serenity',
 'Magnificent Blue Sapphire stone in silver setting.
  A serene ring with timeless elegance.',
 450.00, 8, 'assets/images/blue-1.png',
 'assets/images/blue-2.png', 'assets/images/blue-3.png', 4, 1),
(5, 'Passionate Fire',
 'Vibrant Ruby stones on delicate 18K gold chain.
  A stunning anklet that brings passion and energy to every step.',
 580.00, 20, 'assets/images/ancklet_11.png',
 'assets/images/ankelt_2.png', 'assets/images/ankelt_3.png', 5, 1),
(6, 'Crystal Radiance',
 'Premium sunglasses featuring genuine multi-colored gemstone crystals.
  A sophisticated statement piece for the modern mystic.',
 350.00, 25, 'assets/images/sun_1.png',
 'assets/images/sun_2.png', 'assets/images/sun_3.png', 7, 1),
(7, 'Emerald Essence',
 'Rich Emerald crystal pendant with brilliant diamonds on silver chain.
  A luxurious and sophisticated accessory.',
 2500.00, 10, 'assets/images/FOURTHPICK-1.png',
 'assets/images/FOURTHPICK-2.png', 'assets/images/FOURTHPICK-3.png', 6, NULL),
(8, 'Luminous Pearl',
 'Lustrous Pearl crystals with silver accents.
  A serene and elegant bracelet for everyday grace.',
 850.00, 10, 'assets/images/FIFTHPICK-1.png',
 'assets/images/fifthpick_2.png', 'assets/images/fifthpick_3.png', 1, NULL),
(9, 'Timeless Brilliance',
 'Most precious Diamond on platinum chain.
  A timeless pendant with unmatched brilliance.',
 5000.00, 5, 'assets/images/dimond-1.png',
 'assets/images/dimond-2.png', 'assets/images/dimond-3.png', 6, NULL);
```

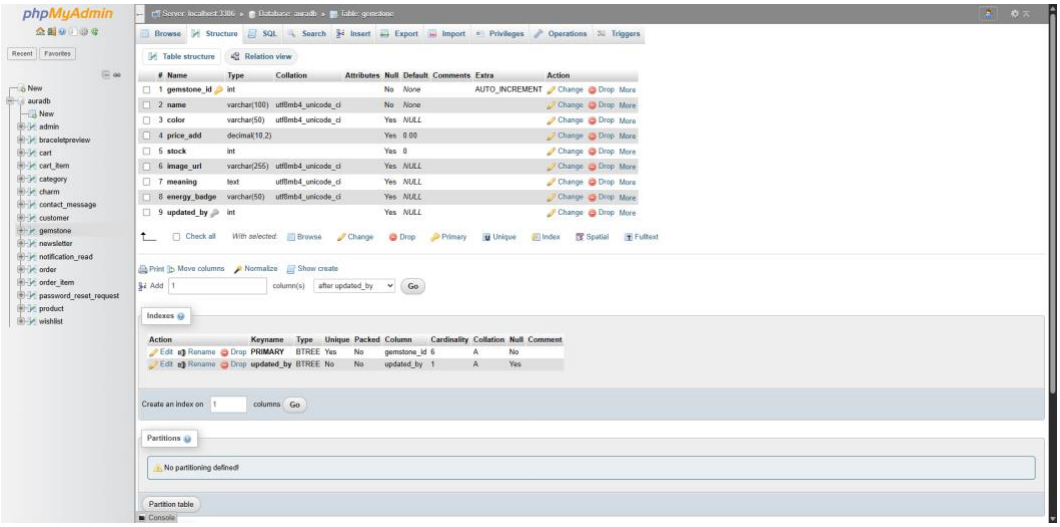


Figure 5: Gemstone table structure

SQL Code:

```
INSERT INTO `gemstone` (`gemstone_id`, `name`, `color`,
  `price_add`, `stock`, `image_url`, `meaning`,
  `energy_badge`, `updated_by`) VALUES
(1, 'Amethyst', 'Purple', 15.00, 100,
 'assets/images/gemstones/amethyst.png',
 'Stone of peace and intuition', 'Calming', 1),
(2, 'Rose Quartz', 'Pink', 12.00, 120,
 'assets/images/gemstones/rose-quartz.png',
 'Stone of universal love', 'Love', 1),
(3, 'Blue Zircon', 'Blue', 20.00, 80,
 'assets/images/gemstones/blue-zircon.png',
 'Stone of wisdom and clarity', 'Clarity', 1),
(4, 'Black Tourmaline', 'Black', 18.00, 90,
 'assets/images/gemstones/black-tourmaline.png',
 'Protective stone', 'Protection', 1),
(5, 'Citrine', 'Yellow', 14.00, 110,
 'assets/images/gemstones/citrine.png',
 'Stone of abundance', 'Energy', 1),
(6, 'Lapis Lazuli', 'Deep Blue', 16.00, 70,
 'assets/images/gemstones/lapis.png',
 'Stone of truth', 'Truth', 1);
```

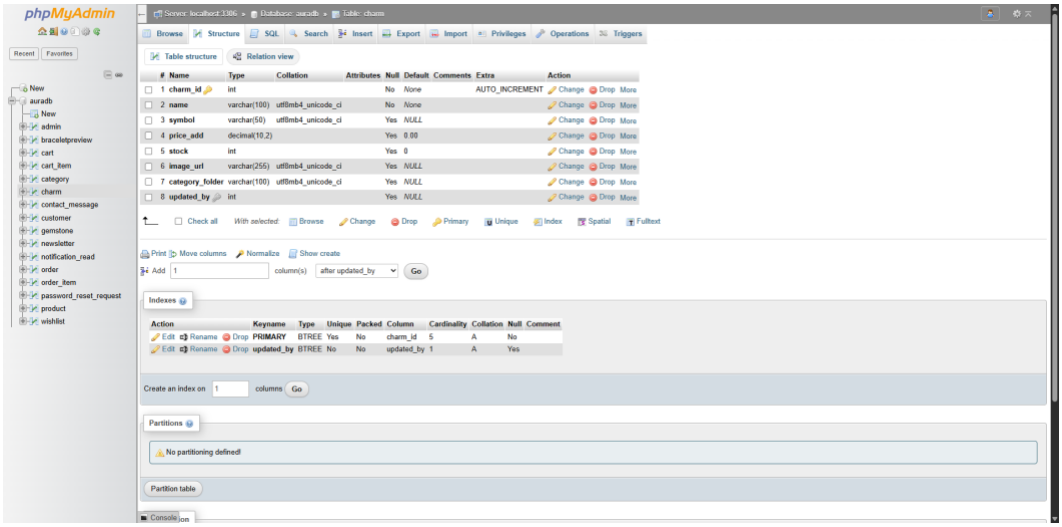


Figure 6: Charm table structure

SQL Code:

```
INSERT INTO `charm` (`charm_id`, `name`, `symbol`,
  `price_add`, `stock`, `image_url`,
  `category_folder`, `updated_by`) VALUES
(1, 'Silver Heart', 'Heart', 10.00, 200,
 'assets/images/charms/heart.png', 'Basic', 1),
(2, 'Star', 'Star', 8.00, 180,
 'assets/images/charms/star.png', 'Basic', 1),
(3, 'Moon', 'Moon', 10.00, 150,
 'assets/images/charms/moon.png', 'Basic', 1),
(4, 'Evil Eye', 'EvilEye', 12.00, 130,
 'assets/images/charms/evileye.png', 'Protection', 1),
(5, 'Butterfly', 'Butterfly', 11.00, 120,
 'assets/images/charms/butterfly.png', 'Nature', 1);
```

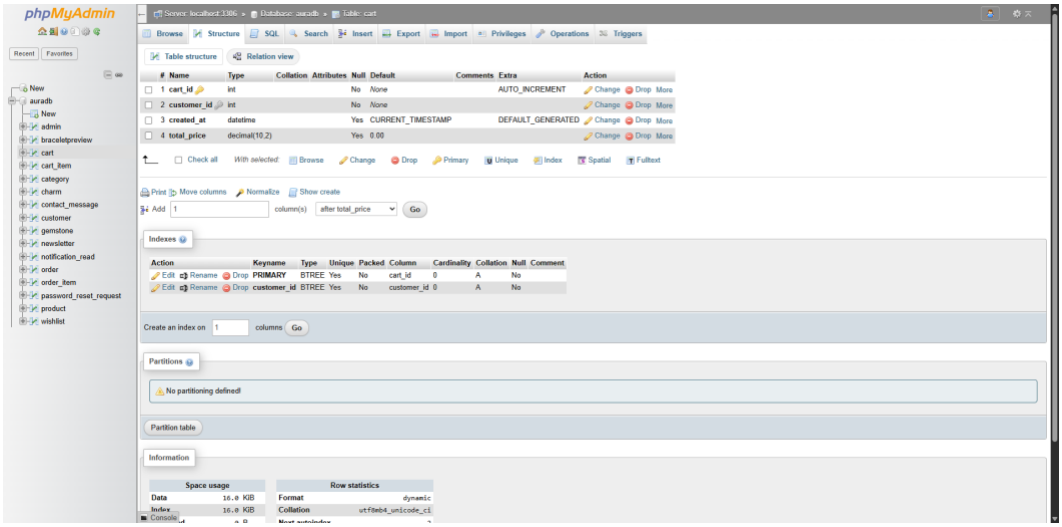


Figure 7: Cart table structure

SQL Code:

```
-- Cart table is transactional (created per customer).
-- Fields: cart_id, customer_id (FK), created_at, total_price
-- A cart is automatically created when a user adds their first item.
```

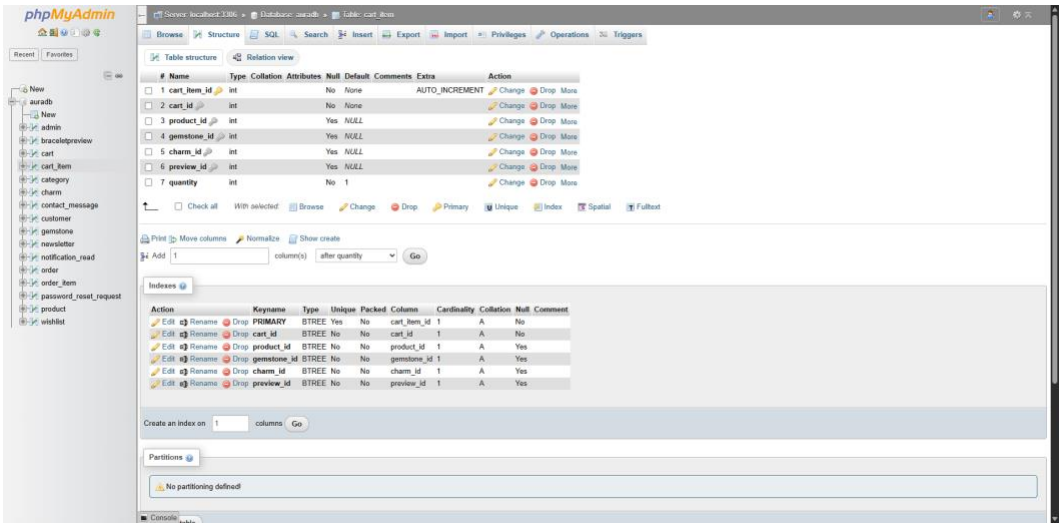


Figure 8: Cart Item table structure

SQL Code:

```
-- Cart Item stores individual products in a cart.
-- Fields: cart_item_id, cart_id (FK), product_id (FK),
--         gemstone_id (FK), charm_id (FK), preview_id (FK), quantity
```

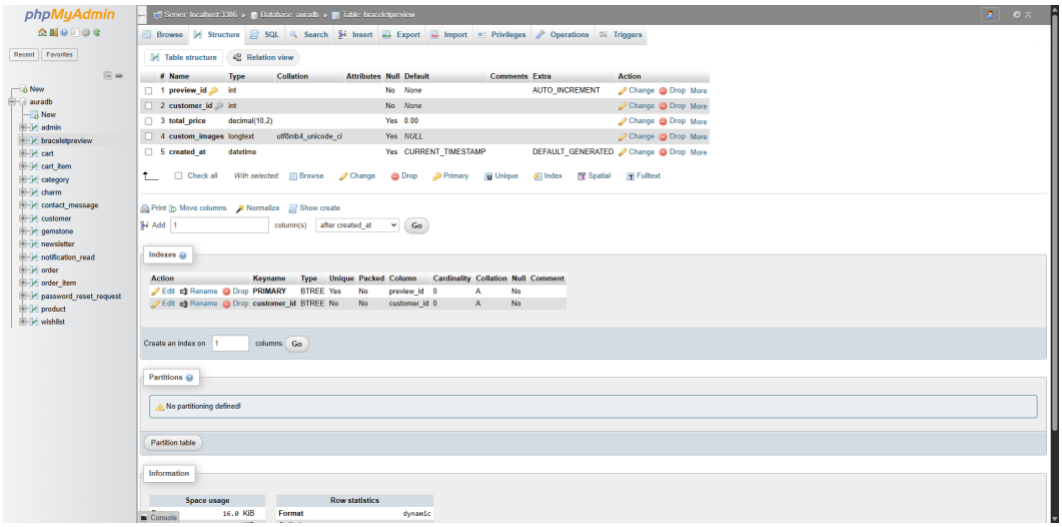


Figure 9: Bracelet Preview table structure

SQL Code:

```
-- Bracelet Preview stores custom bracelet designs.
-- Fields: preview_id, customer_id (FK), total_price,
--         custom_images (JSON), created_at
```

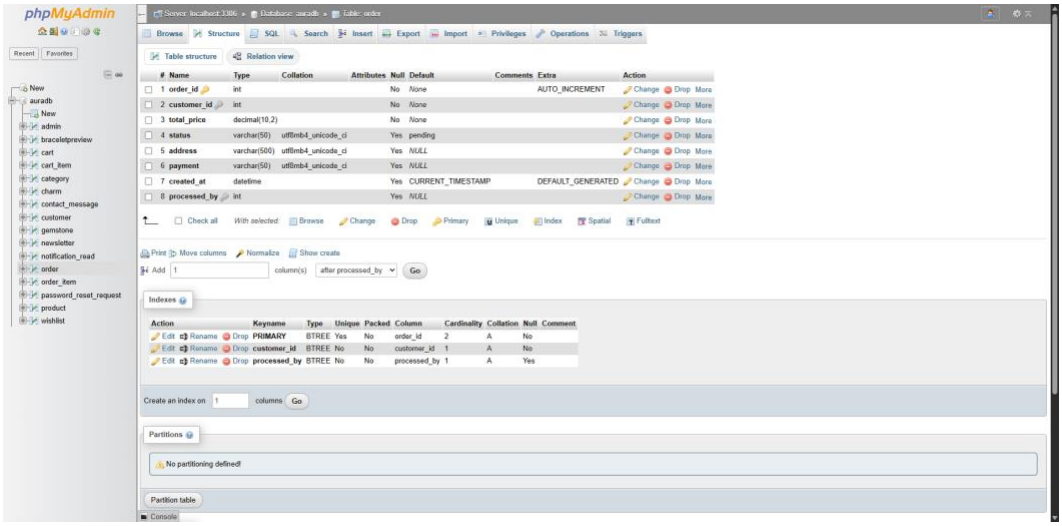


Figure 10: Order table structure

SQL Code:

```
-- Order table stores completed purchases.
-- Fields: order_id, customer_id (FK), total_price,
--         status (pending/processing/completed),
--         address, payment (credit_card/cash_on_delivery),
--         created_at, processed_by (FK to admin)
```

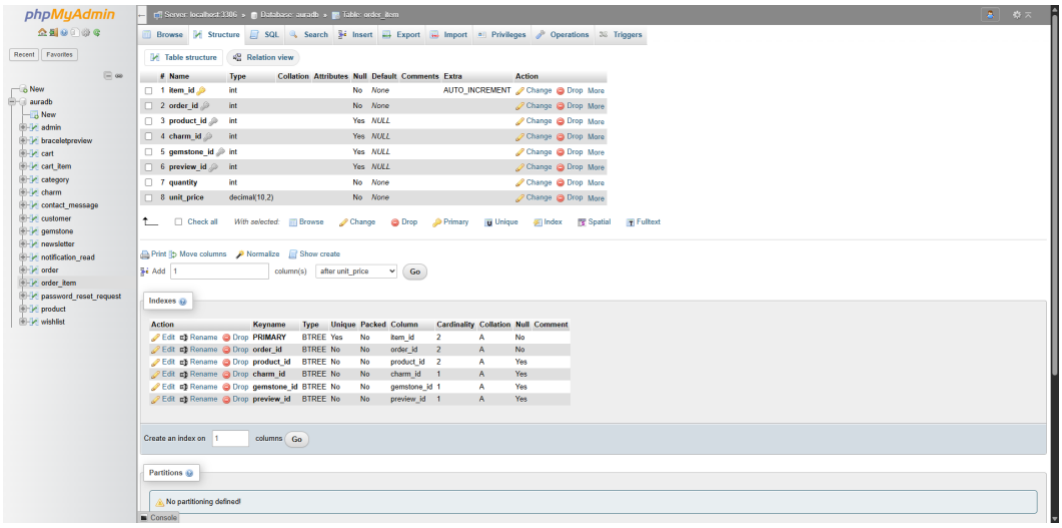


Figure 11: Order Item table structure

SQL Code:

```
-- Order Item stores individual products in an order.
-- Fields: item_id, order_id (FK), product_id (FK),
--         charm_id (FK), gemstone_id (FK), preview_id (FK),
--         quantity, unit_price
```

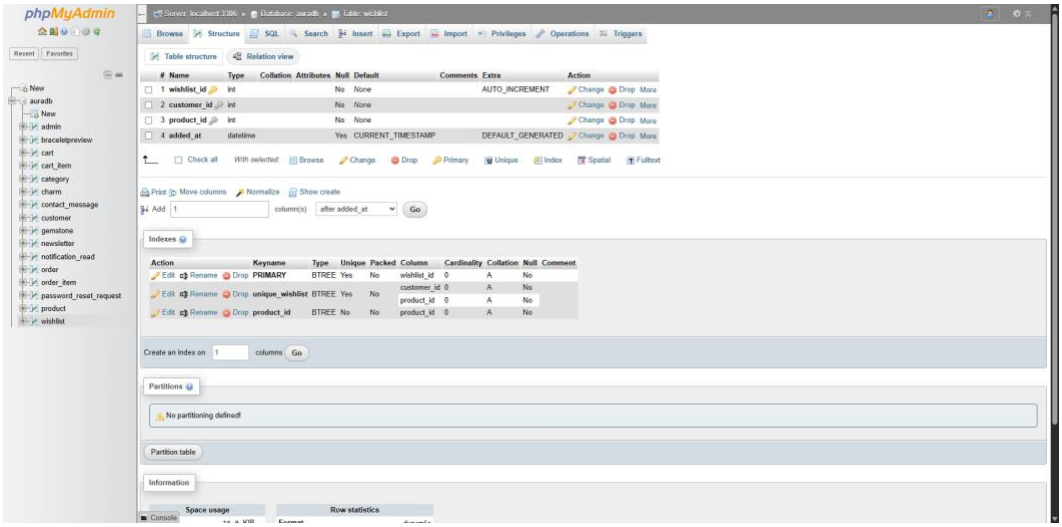


Figure 12: Wishlist table structure

SQL Code:

```
-- Wishlist stores customer product favorites.
-- Fields: wishlist_id, customer_id (FK), product_id (FK), added_at
-- Unique constraint on (customer_id, product_id)
```

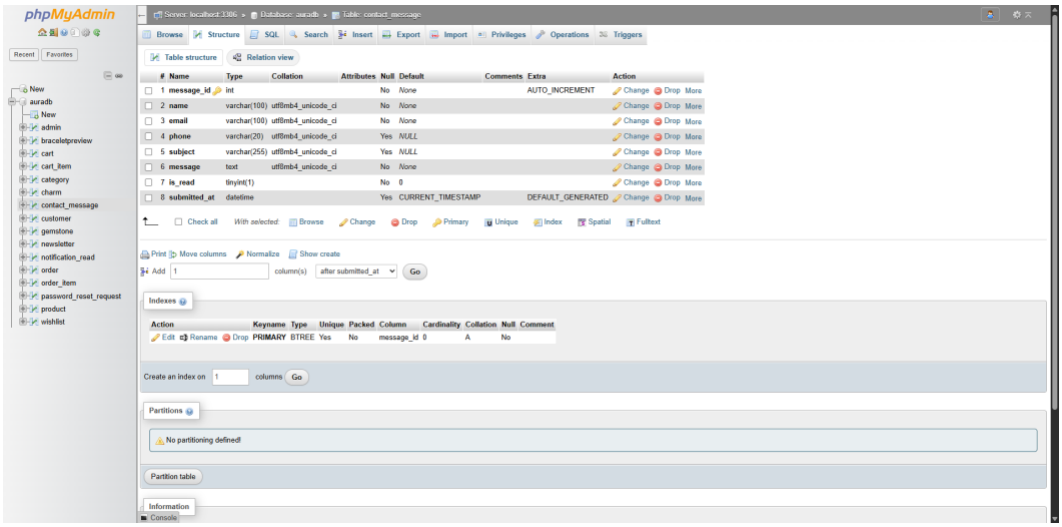


Figure 13: Contact Message table structure

SQL Code:

```
-- Contact Message stores form submissions from the Contact page.
-- Fields: message_id, name, email, phone, subject,
--         message (text), is_read (0/1), submitted_at
```

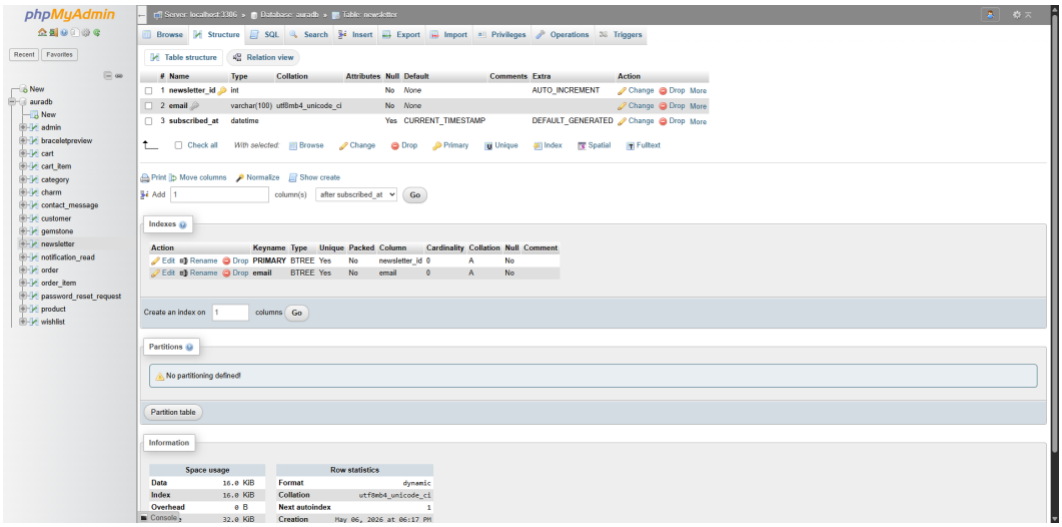


Figure 14: Newsletter table structure

SQL Code:

```
-- Newsletter stores email subscriptions.
-- Fields: newsletter_id, email (unique), subscribed_at
```

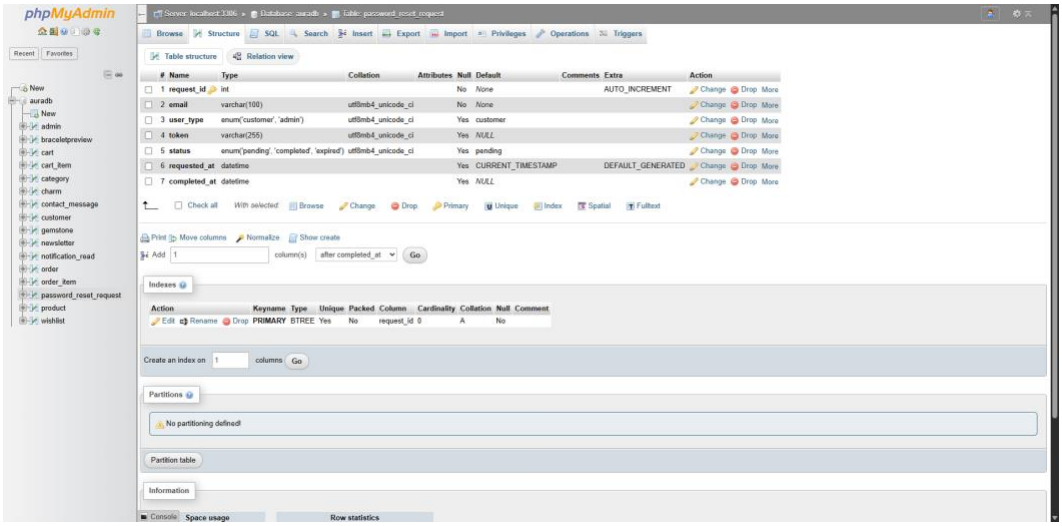


Figure 15: Password Reset Request table structure

SQL Code:

```
-- Password Reset Request tracks forgot-password requests.
-- Fields: request_id, email, user_type (customer/admin),
--         token, status (pending/completed/expired),
--         requested_at, completed_at
```

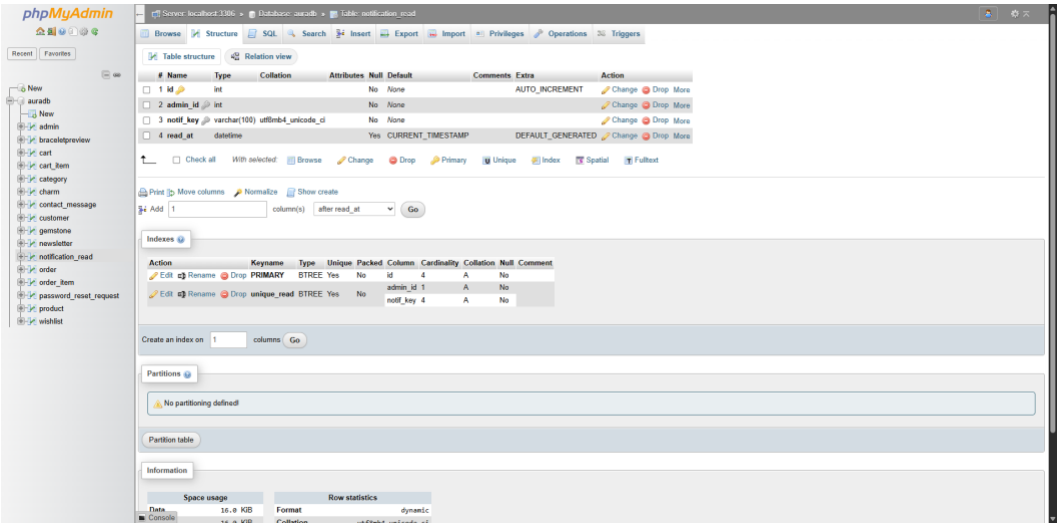


Figure 16: Notification Read table structure

SQL Code:

```
-- Notification Read tracks which admin notifications have been read.
-- Fields: id, admin_id (FK), notif_key, read_at
-- Unique constraint on (admin_id, notif_key)
```

2.2 ER Design

The Entity-Relationship Diagram (ERD) below illustrates the relationships between all database entities in the Aura Artifacts system. The diagram shows primary keys, foreign keys, and the cardinality of each relationship.

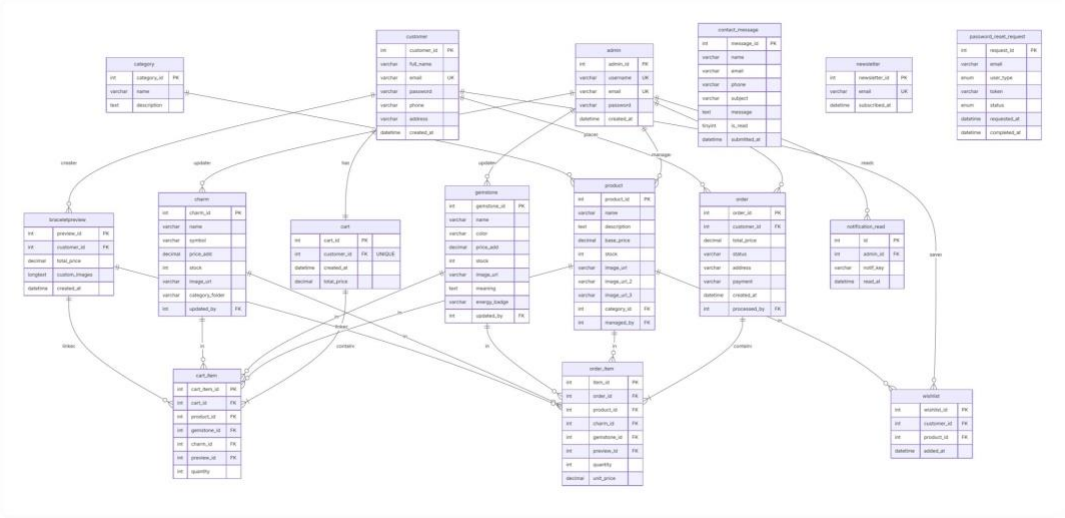


Figure 17: Aura Artifacts Database — Tables Overview

3. Web Application Interface

3.1 Public Pages

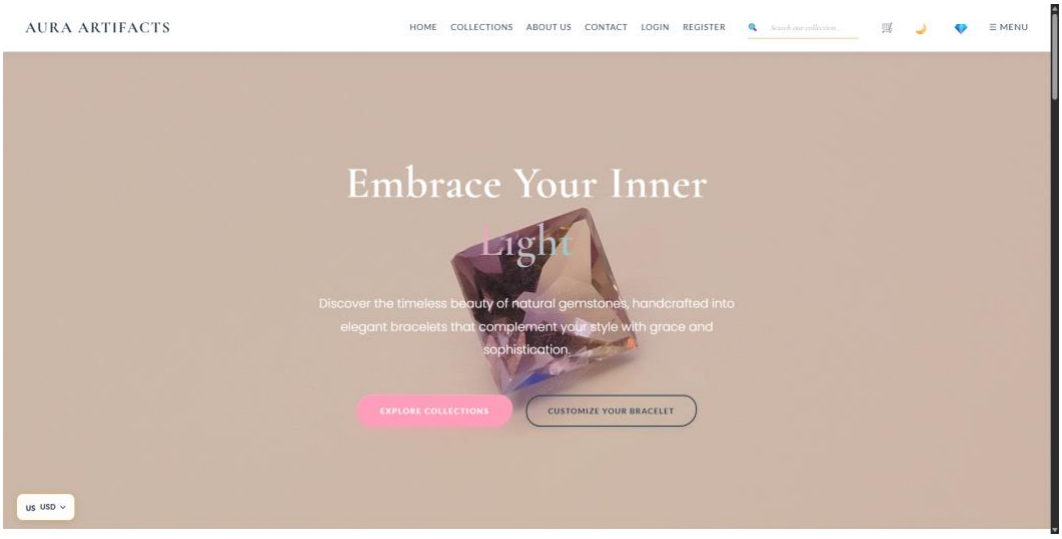


Figure 18: Aura Artifacts Homepage — Hero Section

The homepage features a stunning hero banner showcasing the brand identity with elegant typography and a call-to-action button inviting users to explore the collection.

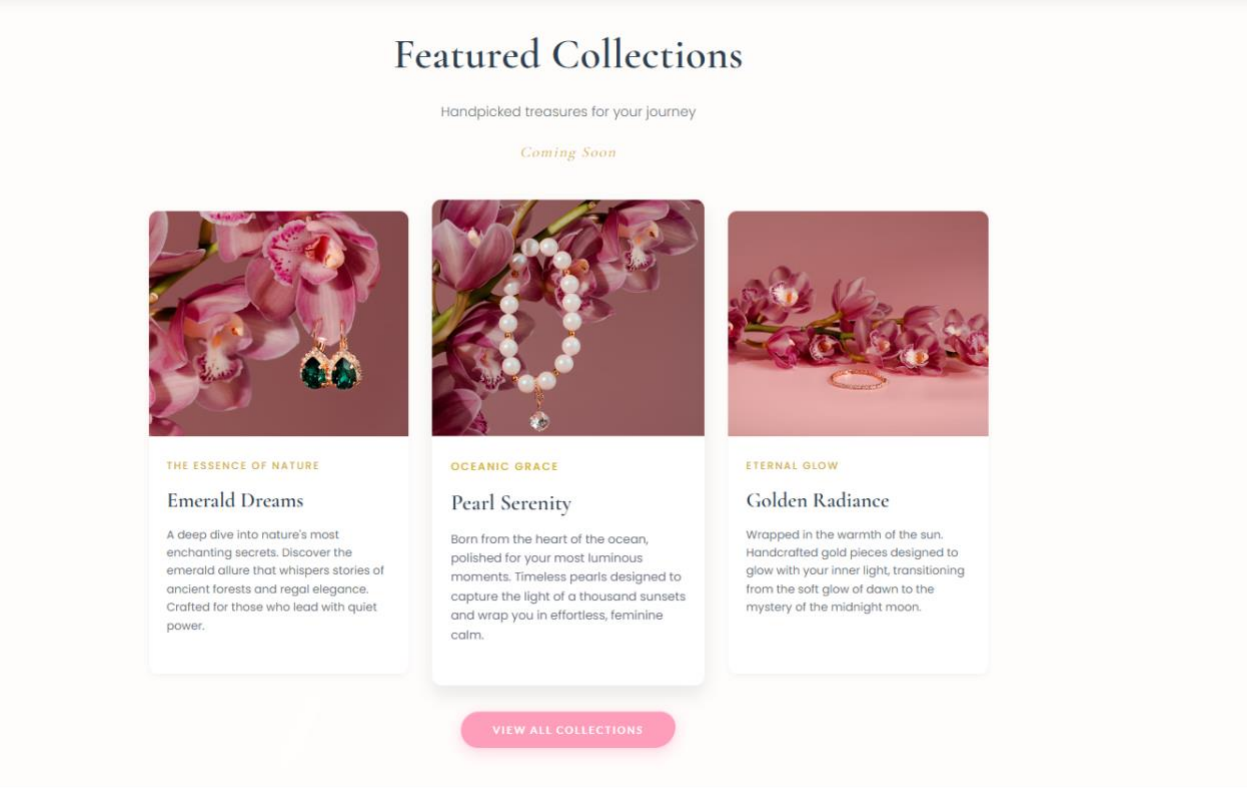


Figure 19: Aura Artifacts Homepage — Featured Products Section

The featured products section displays curated gemstone jewelry pieces with hover effects, pricing, and quick-view options.

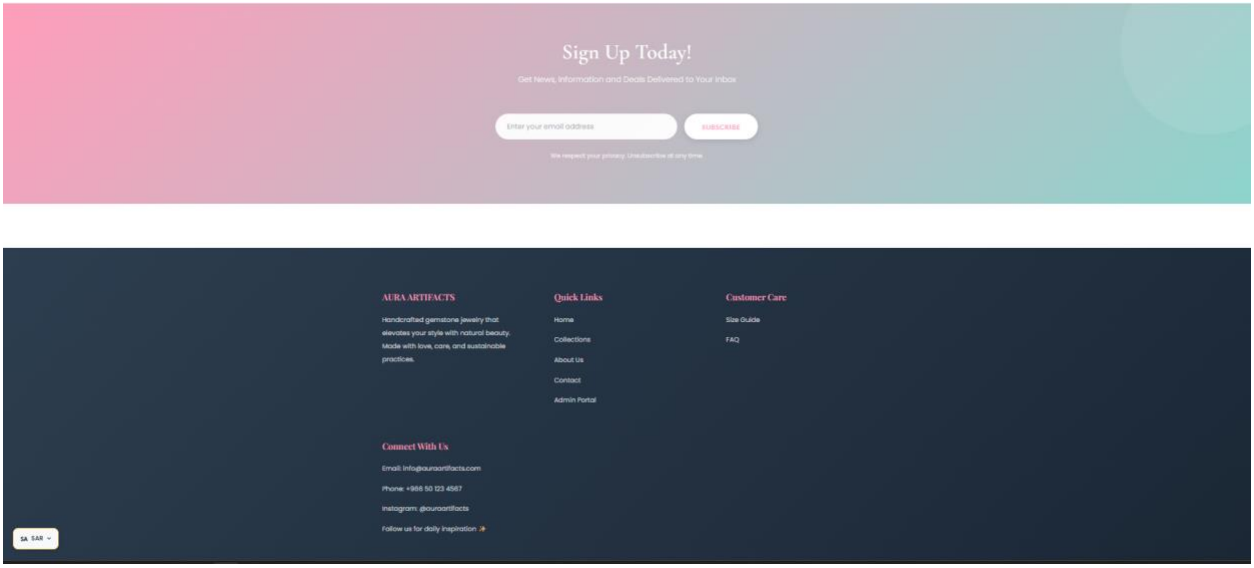


Figure 20: Aura Artifacts Homepage — Footer & Newsletter

The footer includes quick links, customer care information, social media connections, and a newsletter subscription form for email marketing.

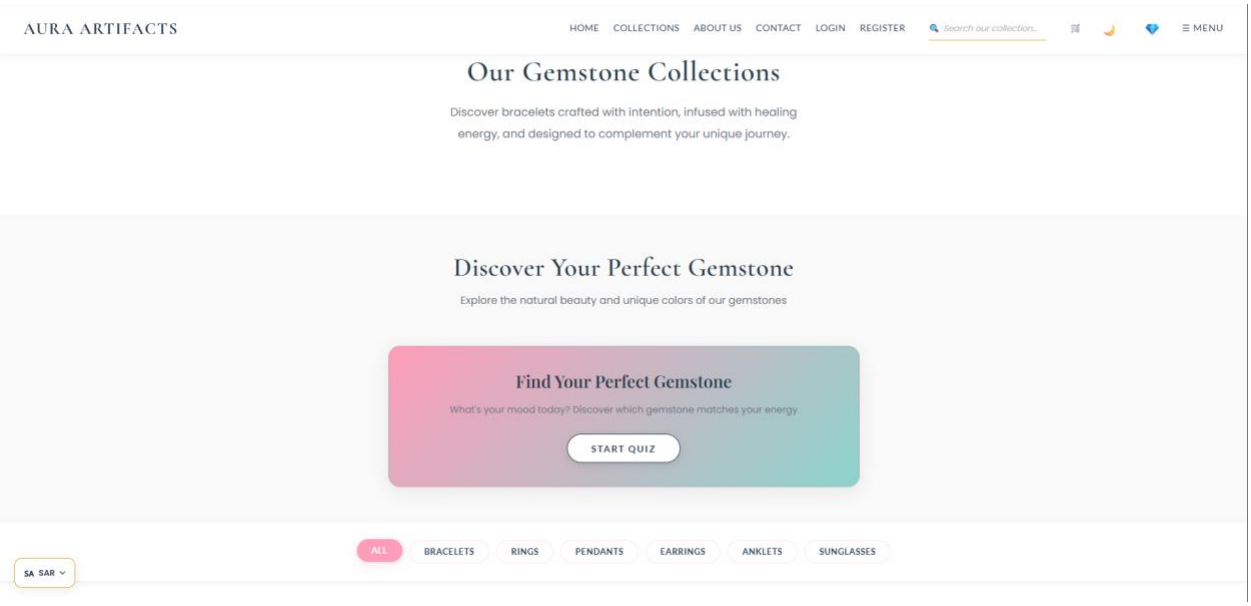


Figure 21: Aura Artifacts Collections Page — Top

The collections page displays all available products organized by category with filtering options. Each product card shows the product image, name, category, and price.

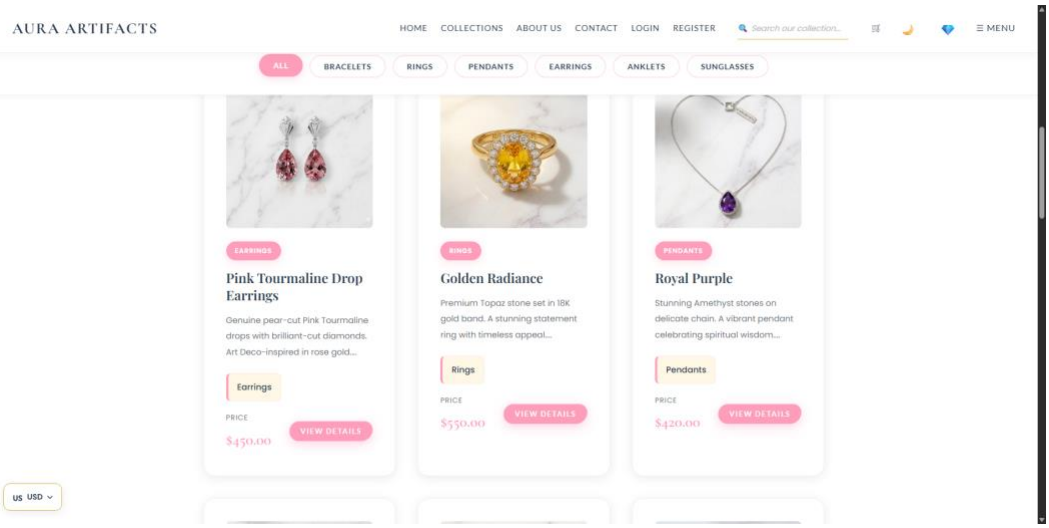


Figure 22: Aura Artifacts Collections Page — Product Grid

Products are displayed in a responsive grid layout with smooth hover animations that reveal additional product images and quick-add functionality.

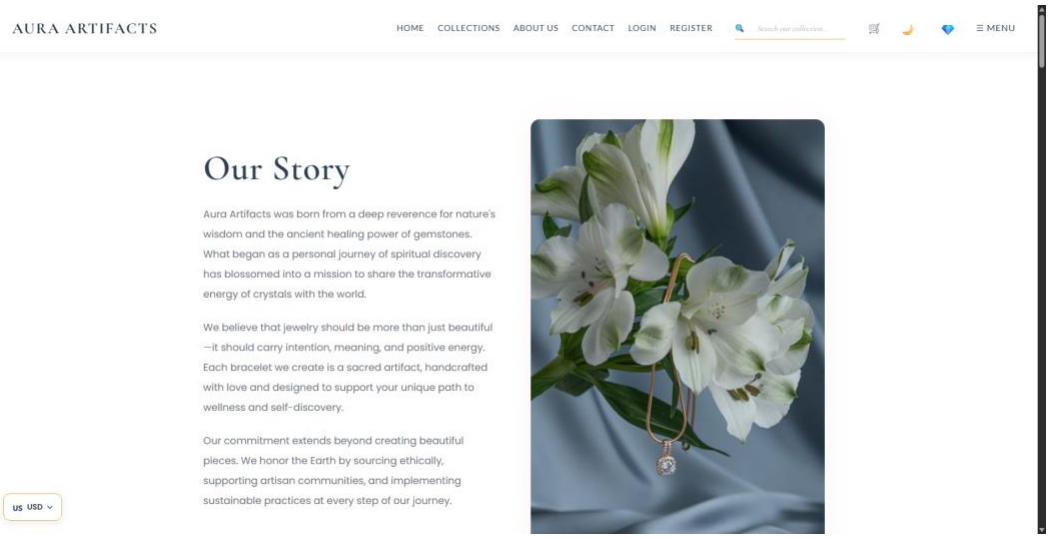


Figure 23: Aura Artifacts About Page — Top

The About page tells the brand story, featuring the company's mission, values, and commitment to handcrafted gemstone jewelry with spiritual significance.

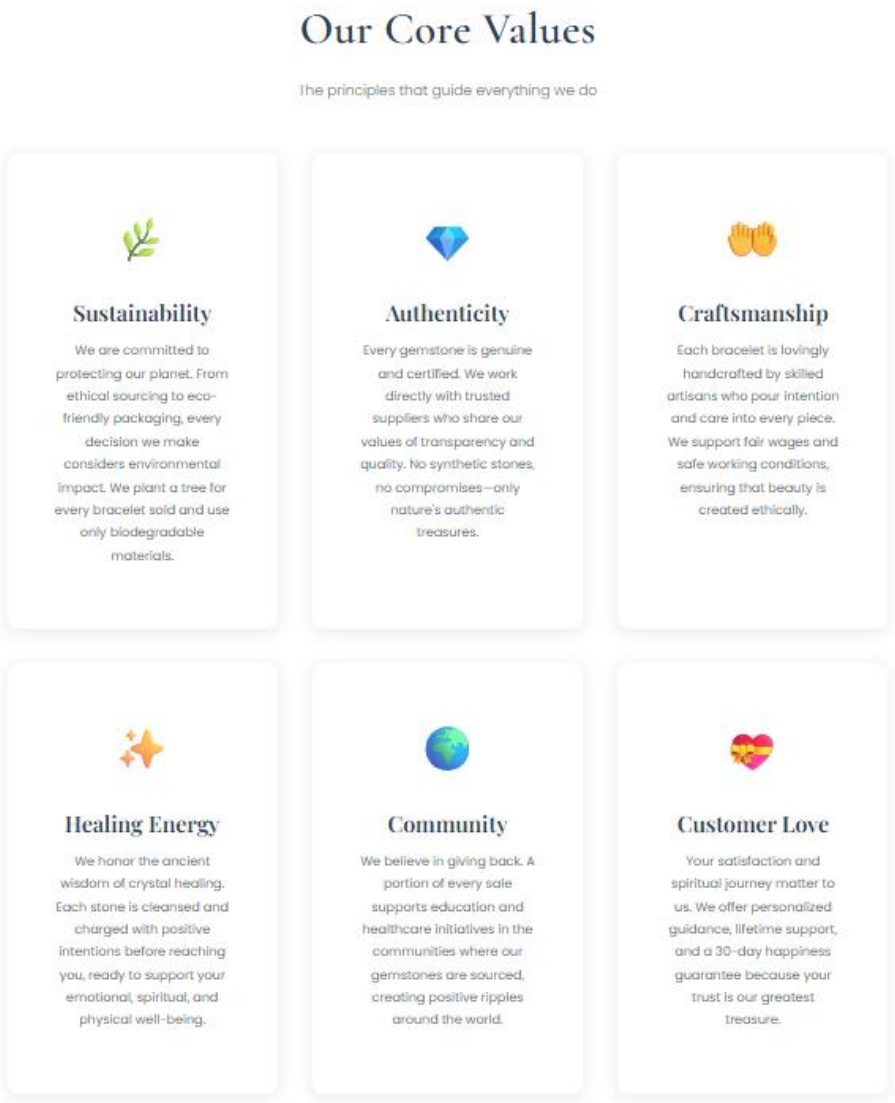


Figure 24: Aura Artifacts About Page — Bottom

The bottom section of the About page highlights the team's expertise and the artisanal process behind each piece.

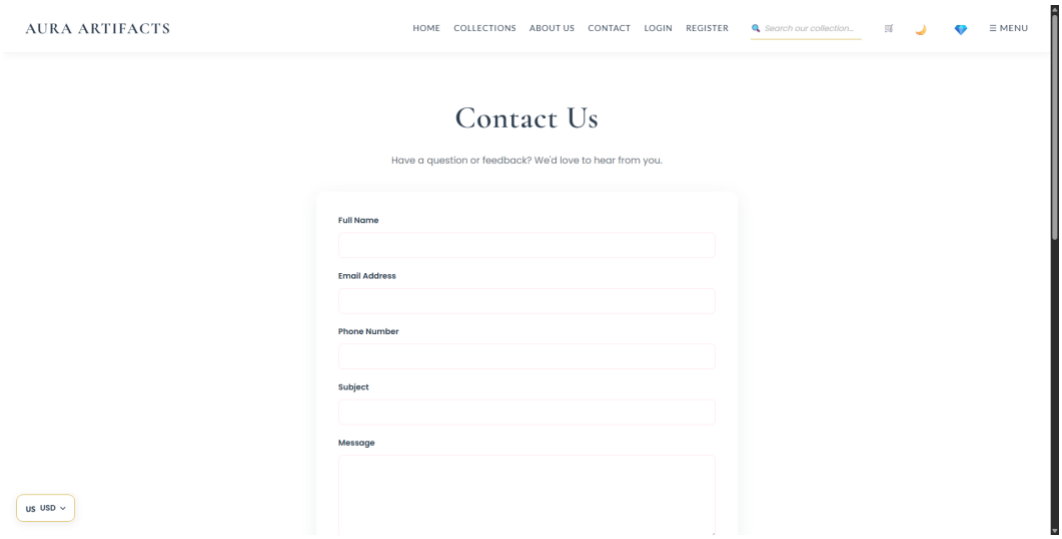


Figure 25: Aura Artifacts Contact Page

The contact page features a contact form with fields for name, email, phone, subject, and message, alongside Google Maps integration showing the store location.

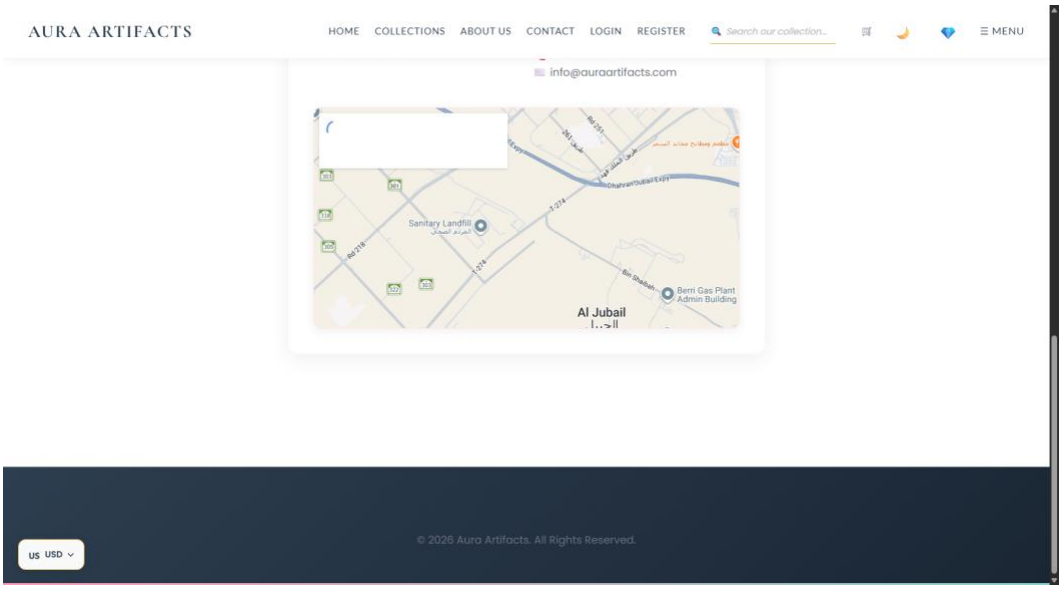


Figure 26: Aura Artifacts Contact Page — Google Maps Integration

The bottom of the contact page displays an embedded Google Map showing the store's physical location in Al Jubail, Saudi Arabia, making it easy for customers to find the store.

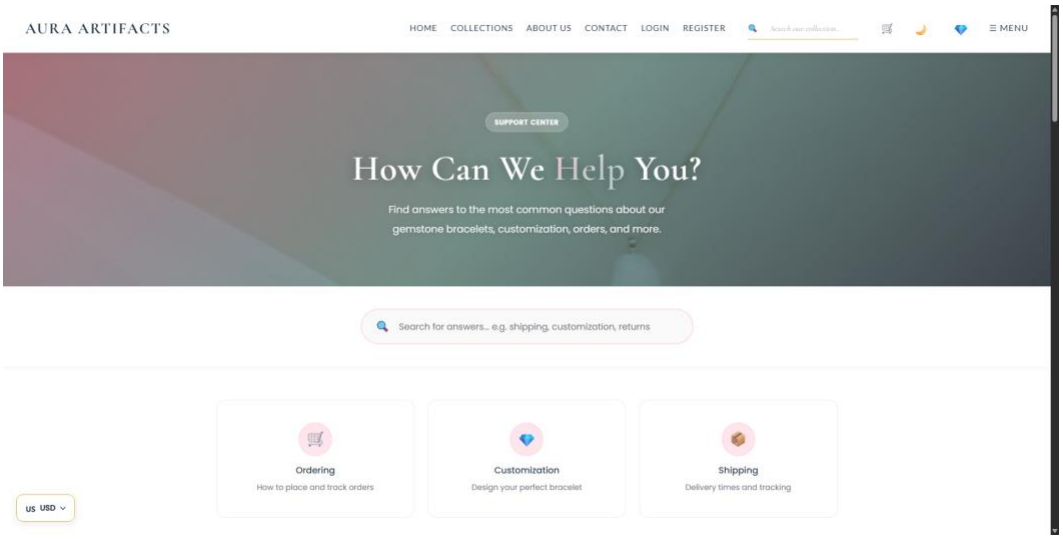


Figure 27: Aura Artifacts Help / FAQ Page — Top

The Help page provides comprehensive FAQ sections covering ordering, shipping, returns, product care, and account management with expandable accordion sections.

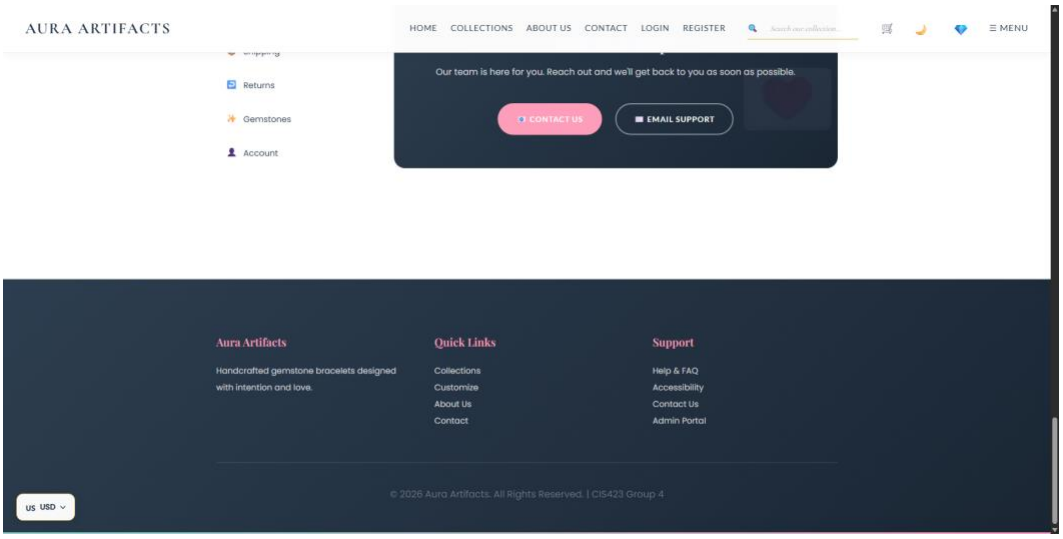


Figure 28: Aura Artifacts Help / FAQ Page — Bottom

Additional help topics and a contact support section for users who need further assistance.

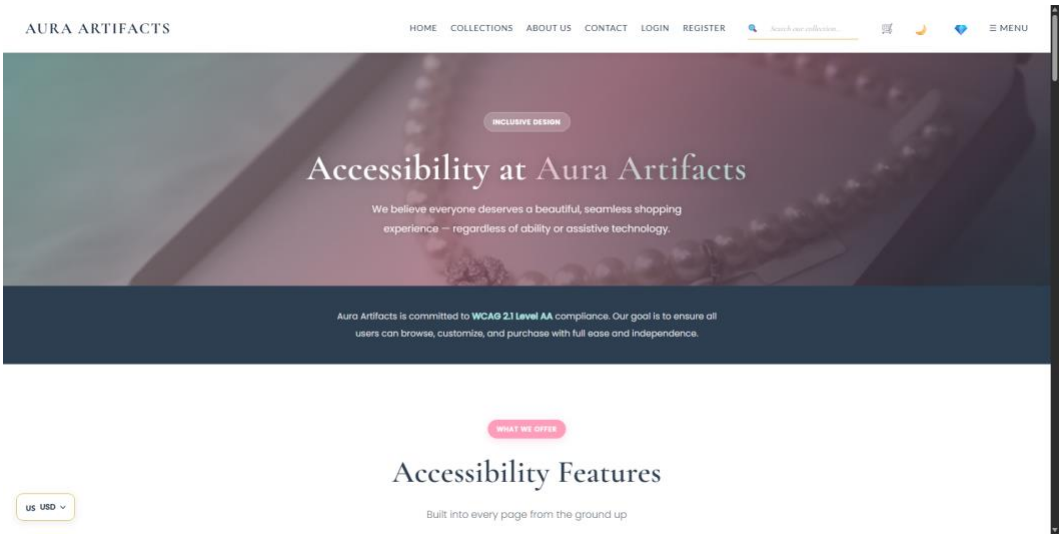


Figure 29: Aura Artifacts Accessibility Page

The Accessibility page demonstrates the platform's commitment to inclusive design, providing information about accessibility features and compliance standards.

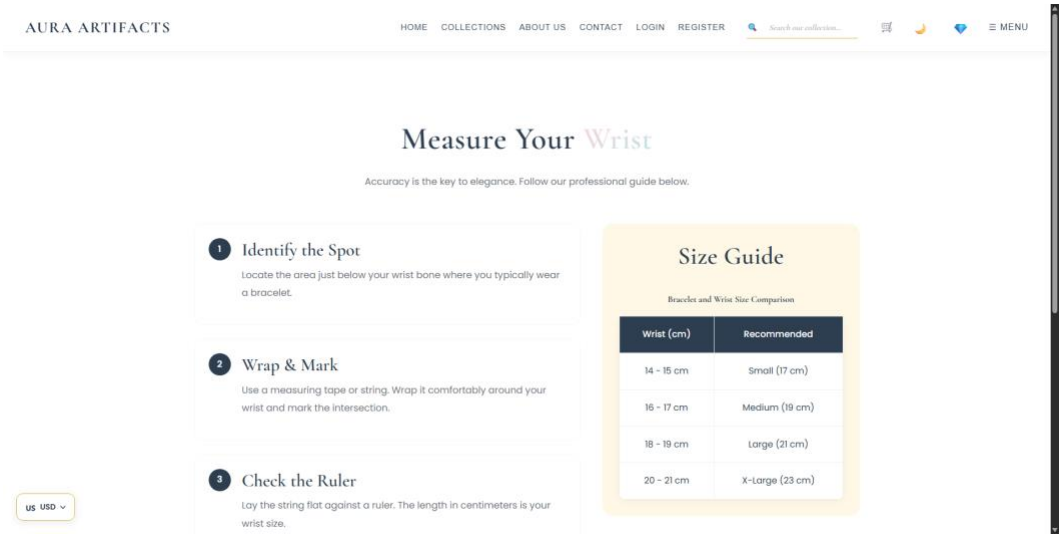


Figure 30: Aura Artifacts Size Guide Page

The Size Guide page helps customers find the perfect fit with detailed measurement charts and visual guides for bracelets, rings, and necklaces.

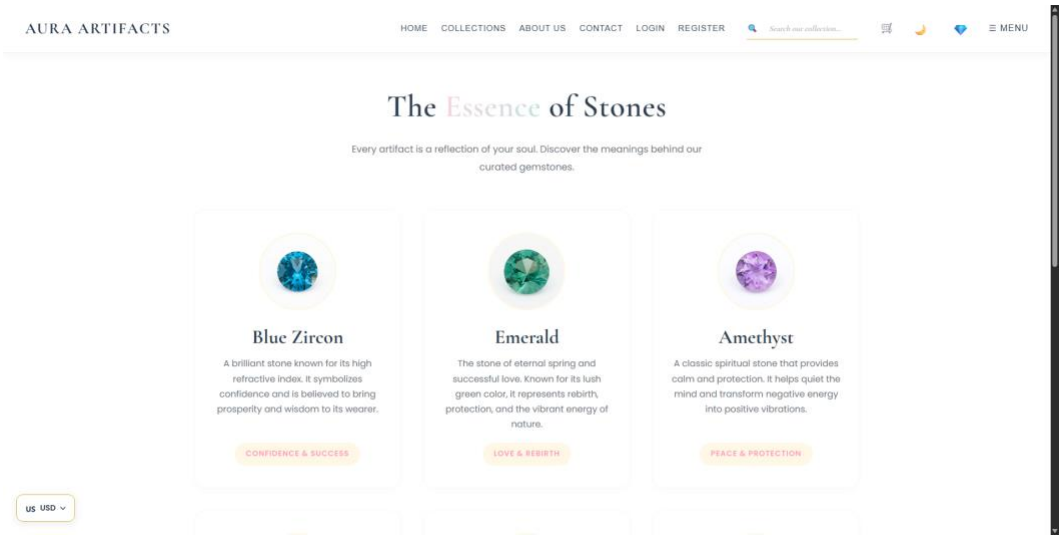


Figure 31: Aura Artifacts Gemstone Meanings Page

The Gemstone Meanings page educates customers about the spiritual properties, healing benefits, and cultural significance of each gemstone used in the jewelry collection.

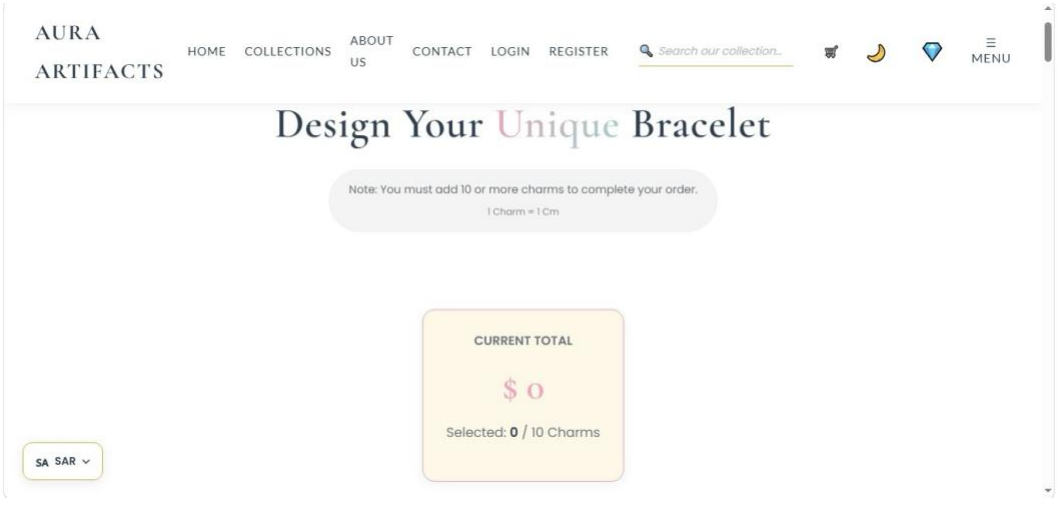


Figure 32: Aura Artifacts Bracelet Customization Page

The Bracelet Customization page allows users to design their own custom bracelet by selecting gemstones and charms, with a real-time preview and price calculator.

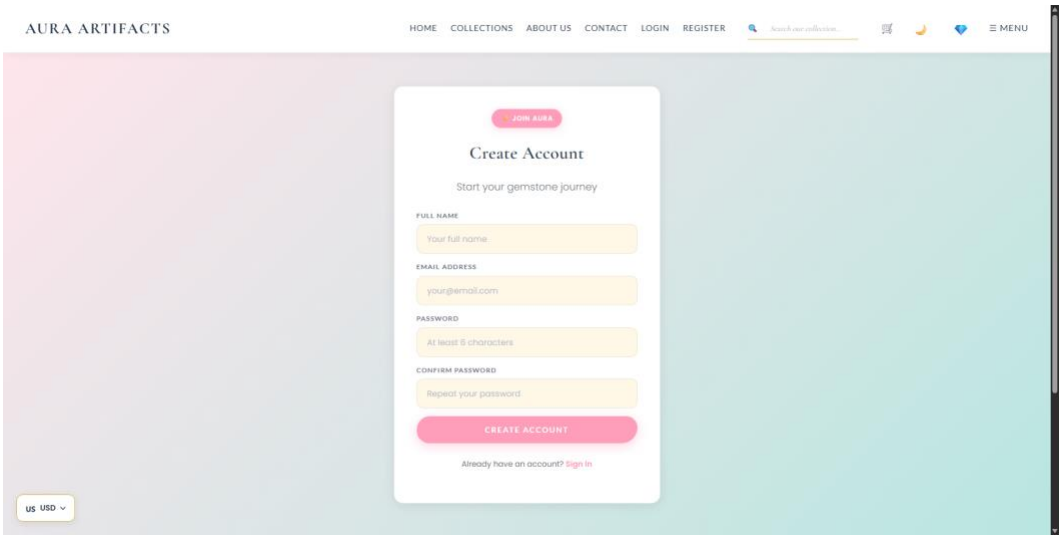


Figure 33: Aura Artifacts Create Account Page

The registration page features a clean form with fields for full name, email, phone, password, and password confirmation. Client-side validation ensures proper email format and password strength.

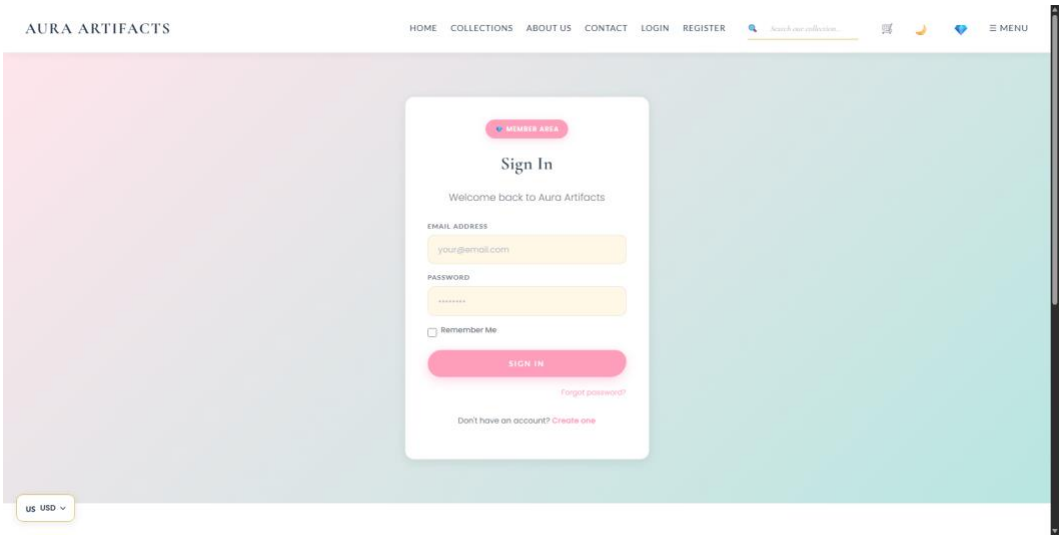


Figure 34: Aura Artifacts Login Page

The login page provides secure authentication with email and password fields, a 'Remember Me' checkbox for persistent sessions (7-day token-based), and links to forgot password and registration.

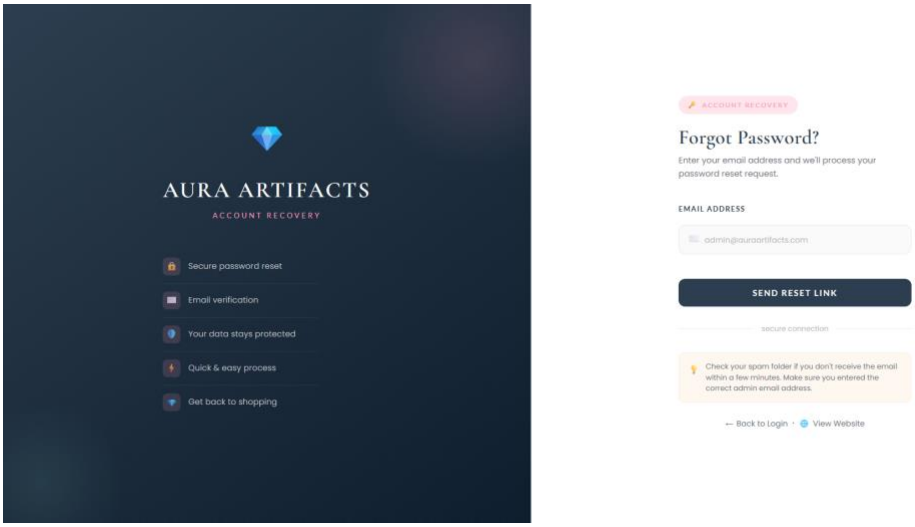


Figure 35: Aura Artifacts Forgot Password Page

The forgot password page allows users to request a password reset by entering their registered email address.

3.2 User Pages (Logged In)

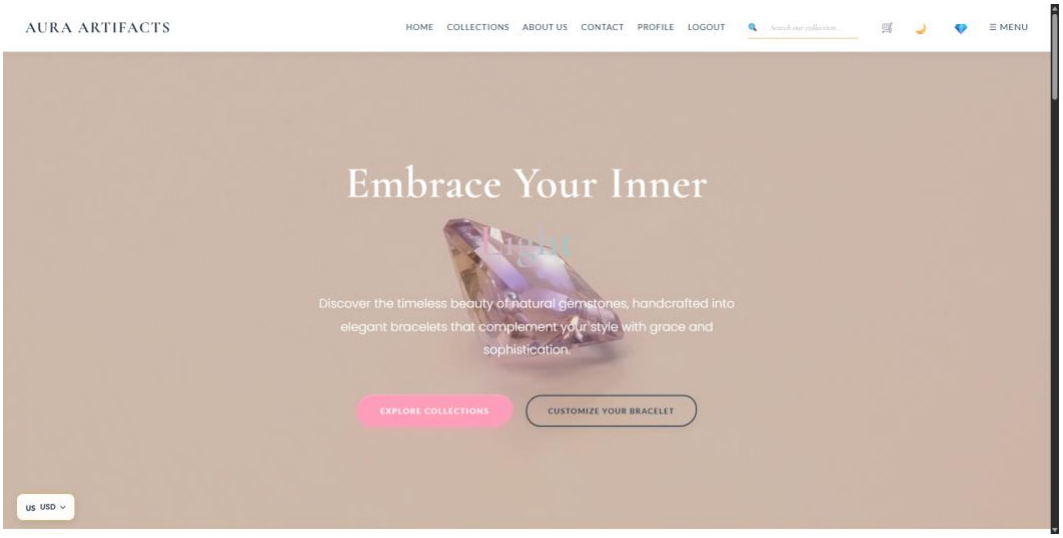


Figure 36: Aura Artifacts Homepage — Logged In User View

After login, the homepage displays personalized navigation with the Profile, Wishlist access, and cart icon with item count.

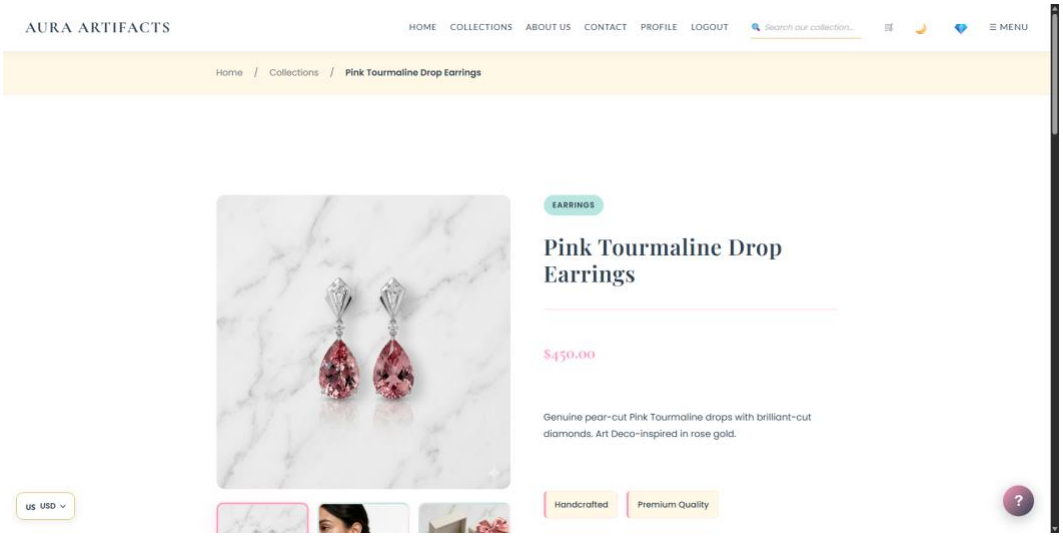


Figure 37: Aura Artifacts Product Detail Page

The product detail page shows multiple product images with a gallery viewer, detailed description, pricing, stock availability, and an 'Add to Cart' button. A help popup provides additional product information.

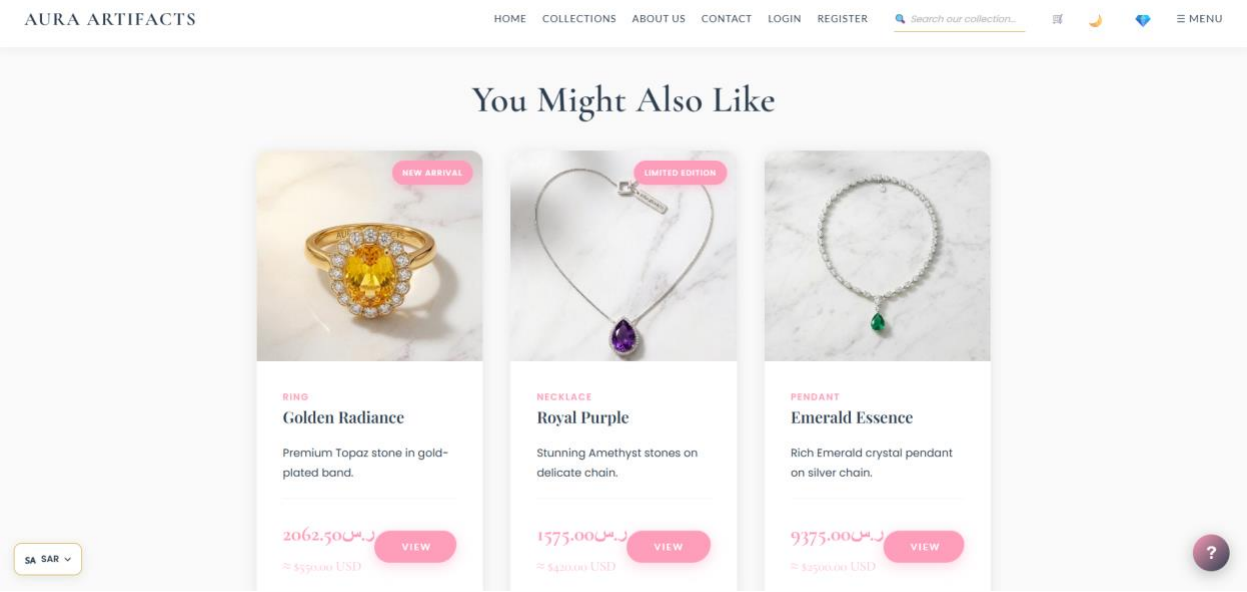


Figure 38: Aura Artifacts Product Detail — Bottom Section

The bottom of the product detail page features related products and additional product information.

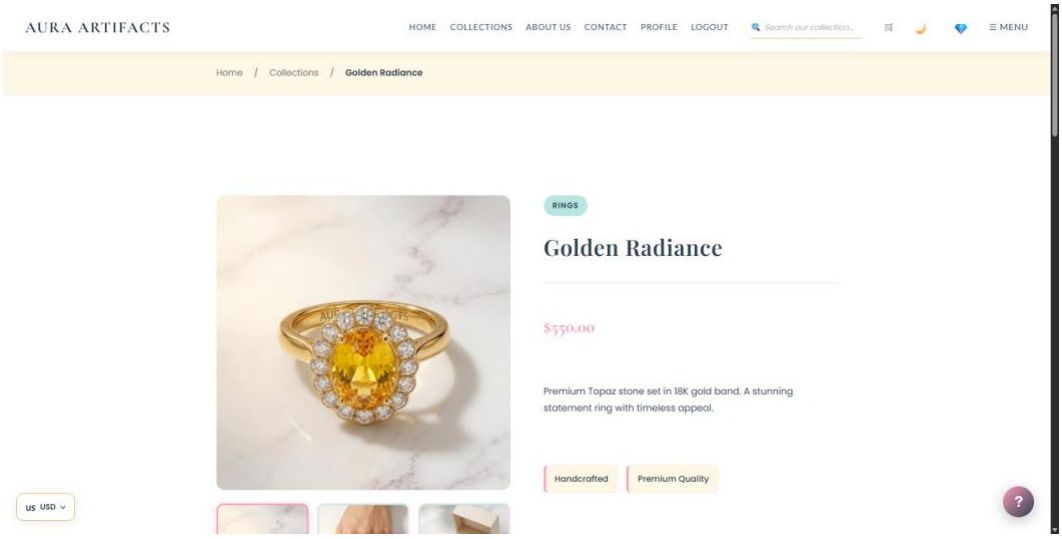


Figure 39: Aura Artifacts Product Detail Page #2

Another product detail view showcasing the consistent design across different product types.

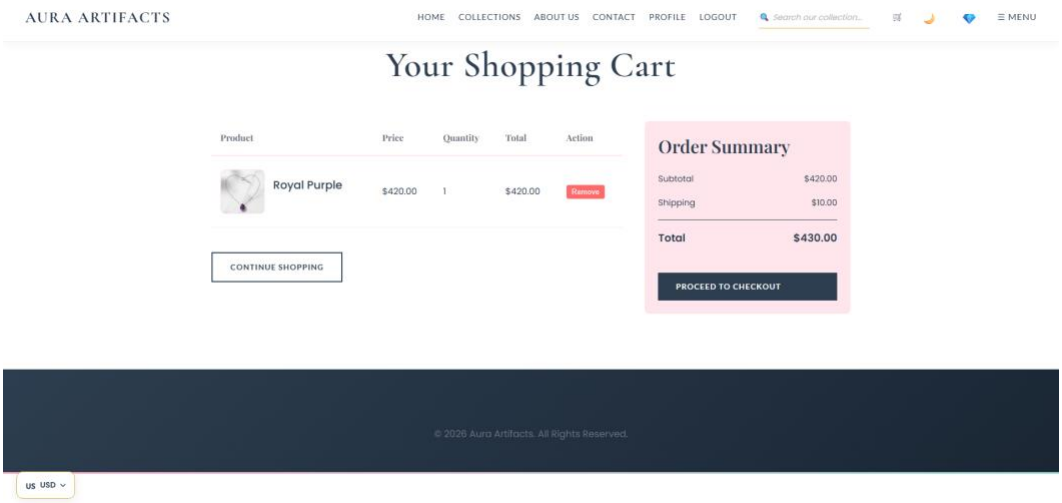


Figure 40: Aura Artifacts Shopping Cart Page

The shopping cart displays all added items with product images, names, prices, quantity controls, and subtotal calculations. Users can update quantities or remove items.

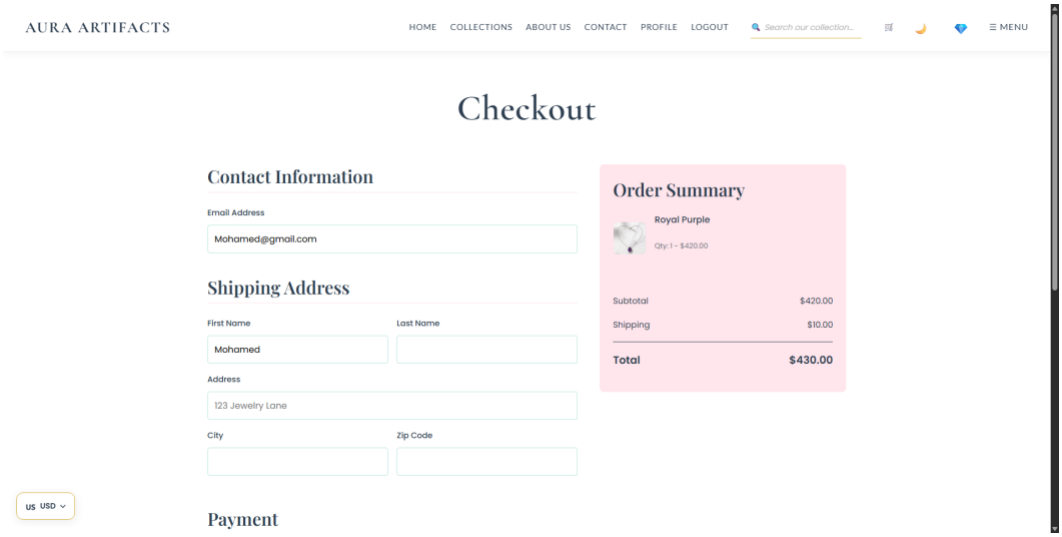


Figure 41: Aura Artifacts Checkout Page — Top

The checkout page collects shipping address information with form validation to ensure all required fields are completed before proceeding to payment.

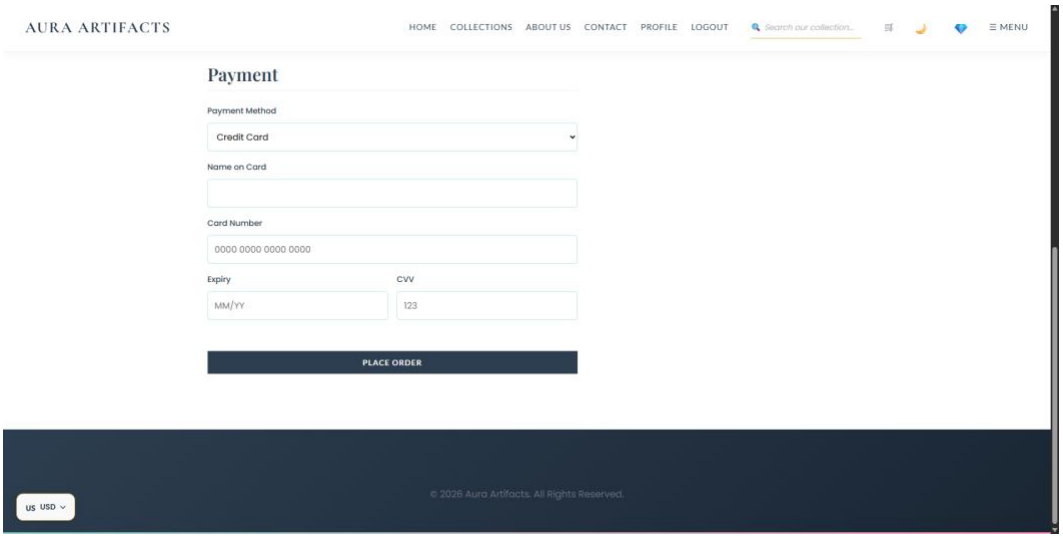


Figure 42: Aura Artifacts Checkout Page — Payment Section

The payment section requires credit card details including card number, expiry date, and CVV. The order summary shows itemized totals and the final amount.

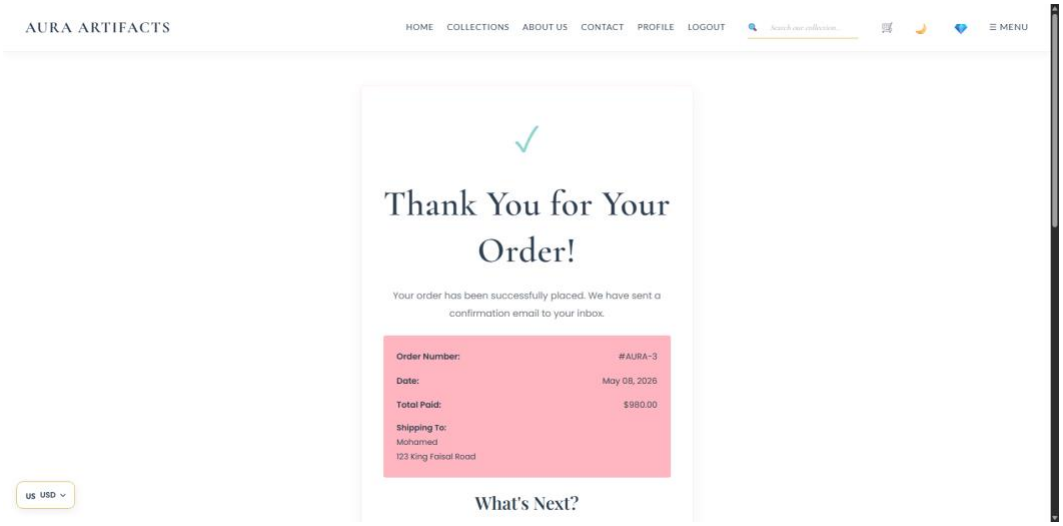


Figure 43: Aura Artifacts Order Confirmation Page

After successfully placing an order, the user is redirected to the order confirmation page displaying the order number, date, total paid, and shipping address. A confirmation email notification is also sent.

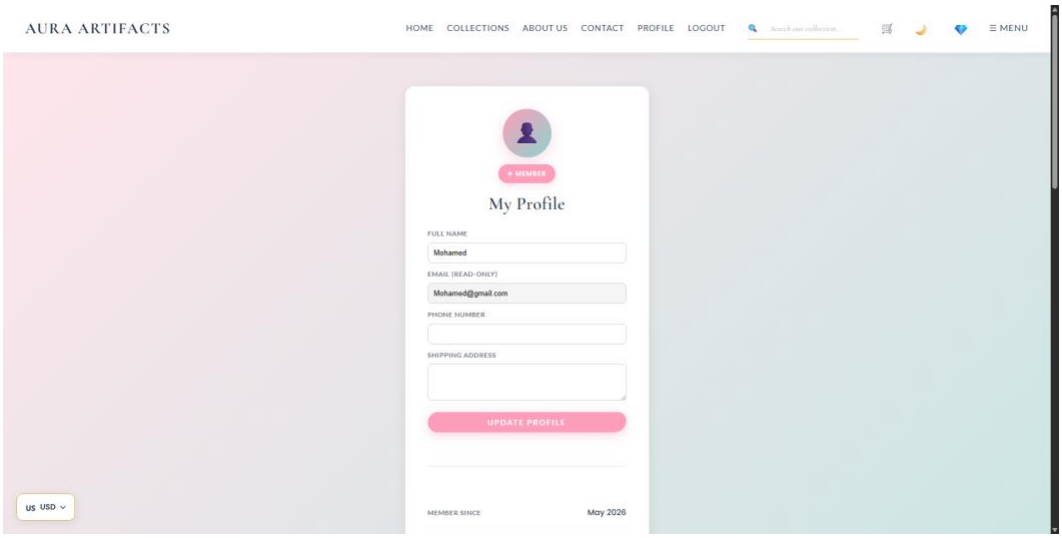


Figure 44: Aura Artifacts User Profile Page

The profile page allows users to view and update their personal information including name, email, phone number, and address.

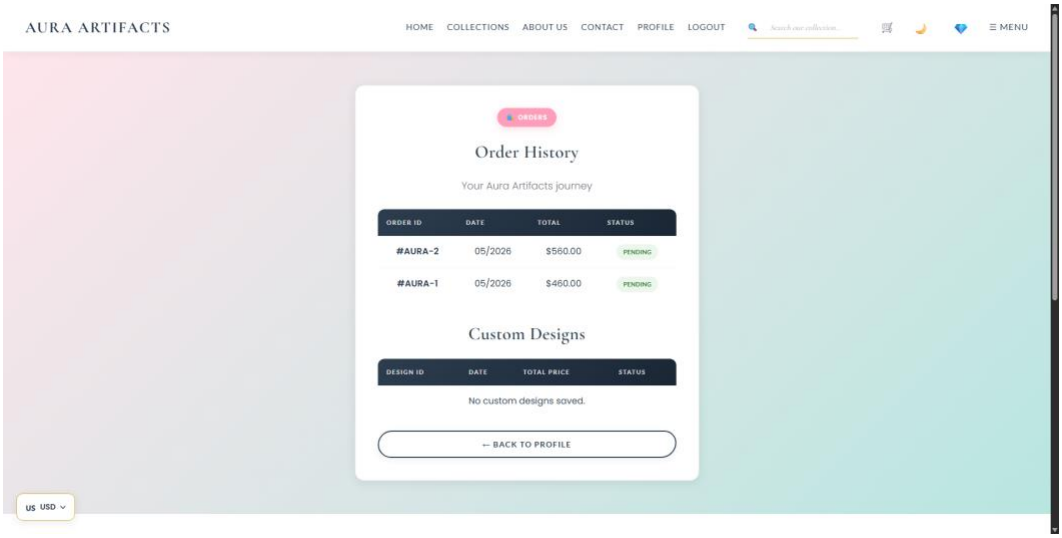


Figure 45: Aura Artifacts Order History Page

The order history page displays all past orders with order numbers, dates, statuses (pending, processing, completed), total amounts, and detailed item breakdowns.

3.2.1 Search Feature

The platform includes a real-time product search feature accessible from the navigation bar on every page. Users can search for products by name or category, with results displayed in an interactive dropdown.

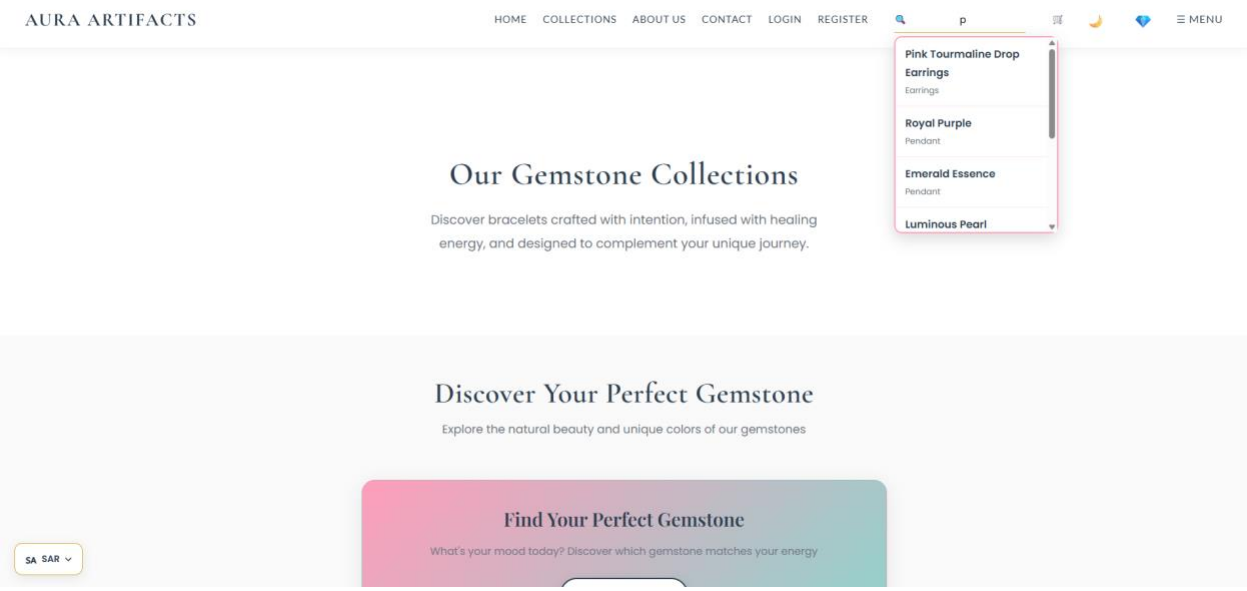


Figure 46: Aura Artifacts Search Feature — Results Dropdown

The search bar filters products in real-time as the user types, showing matching product names and categories in an elegant dropdown.

3.2.2 Wishlist Feature

Logged-in users can add products to their wishlist by clicking the gem icon. The wishlist dropdown shows saved products with images, names, and prices, with quick links to product details and a remove option.

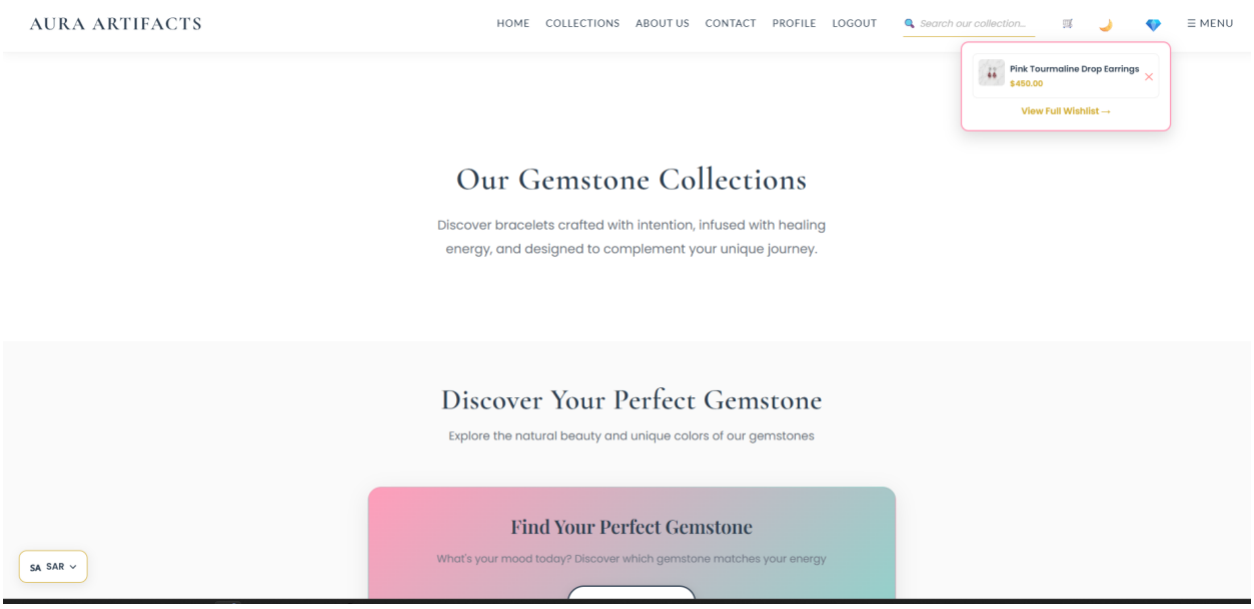


Figure 47: Aura Artifacts Wishlist — Dropdown View

The wishlist dropdown displays the user's saved products with product images, names, and prices. Users can remove items or navigate to the full wishlist in their profile.

3.2.3 Currency Converter (API Integration)

The platform integrates with an external exchange rate API (open.er-api.com) to provide real-time currency conversion. Users can switch between 150+ currencies using a searchable dropdown widget positioned at the bottom-left of every page. The conversion rates are cached locally for performance and updated every hour.

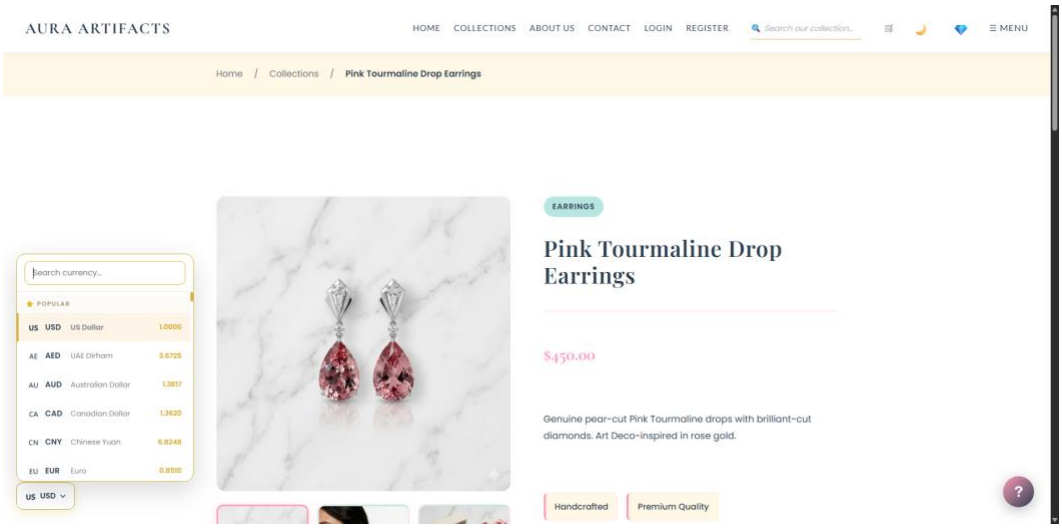


Figure 48: Aura Artifacts Currency Converter — Dropdown Open

The currency converter widget shows popular currencies at the top and a searchable list of 150+ currencies with country flags, currency codes, and live exchange rates.

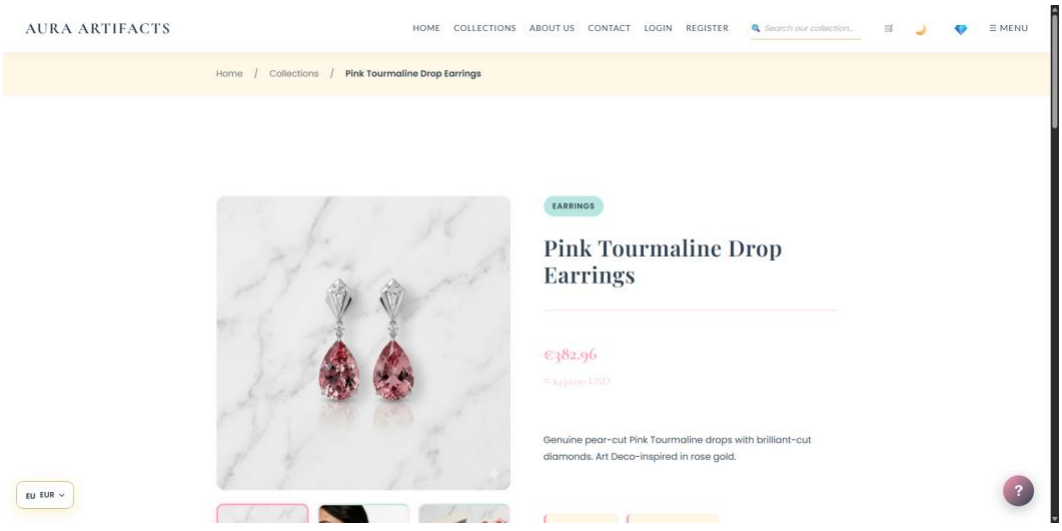


Figure 49: Aura Artifacts Currency Converter — Prices in EUR

After selecting EUR, all product prices on the page are dynamically converted and displayed with the original USD price shown below for reference.

3.2.4 Contact Form Submission

The contact form allows users to send messages directly to the store administrators. Messages are stored in the database and appear in the admin dashboard for management.

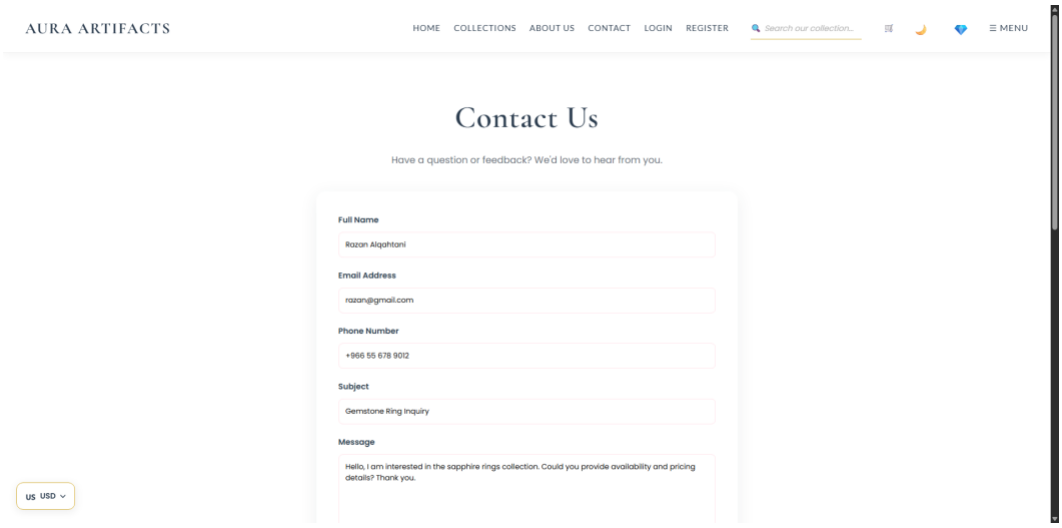


Figure 50: Aura Artifacts Contact Form — Filled Out

The contact form filled with customer details including name, email, phone, subject, and a detailed message. The form includes Google Maps integration showing the store location.

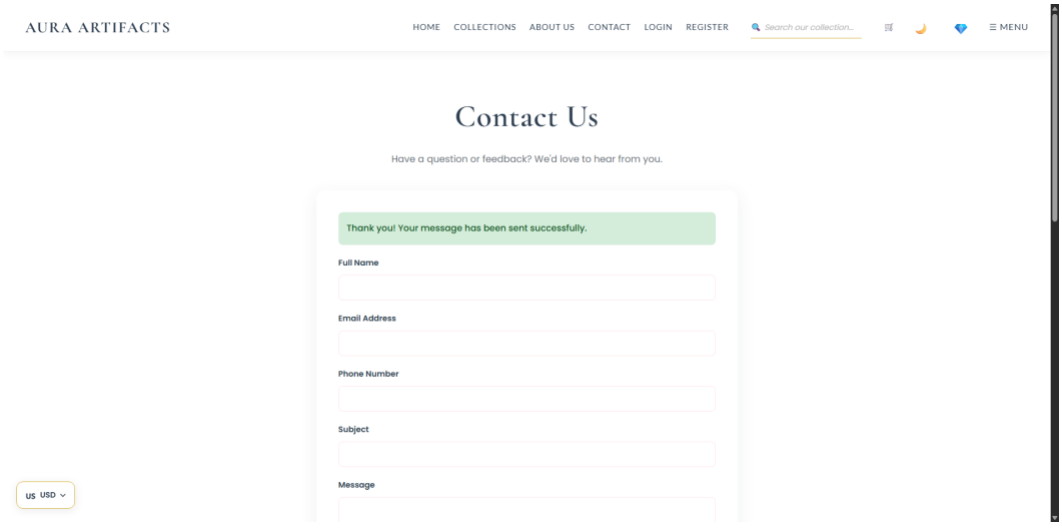


Figure 51: Aura Artifacts Contact Form — Success Message

After successful submission, a confirmation message is displayed and the message is stored in the database for admin review.

3.2.5 Newsletter Subscription

The platform offers a newsletter subscription feature in the footer of the homepage. Users can enter their email address to subscribe for updates about new products and promotions.

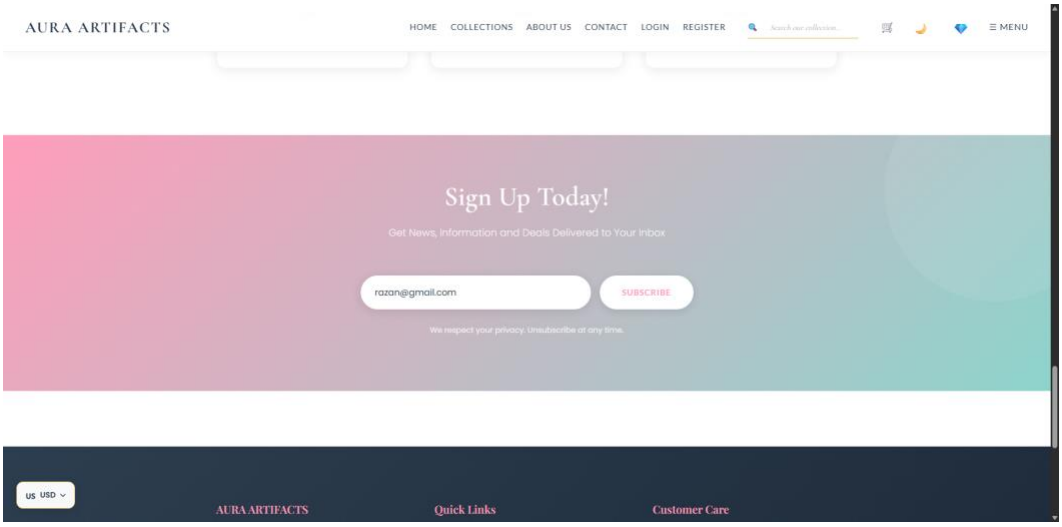


Figure 52: Aura Artifacts Newsletter — Email Entered

The newsletter subscription section in the homepage footer allows users to enter their email for updates and special offers.

3.3 Admin Panel

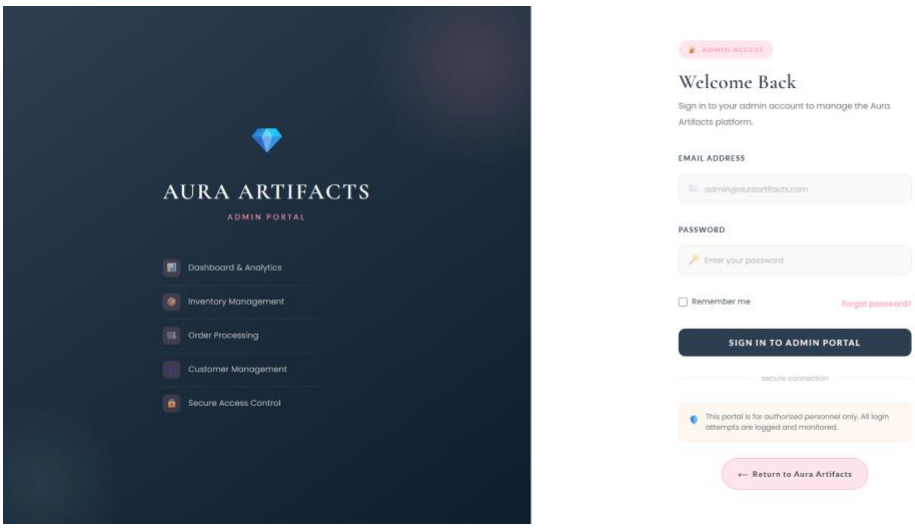


Figure 53: Aura Artifacts Admin Login Page

The admin login page provides a separate, secure authentication portal for administrators with email and password fields.

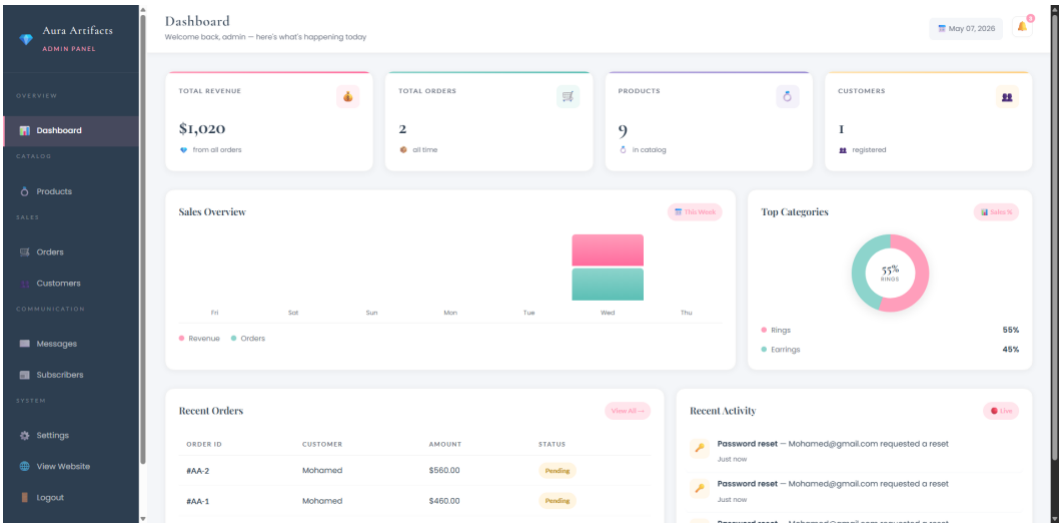


Figure 54: Aura Artifacts Admin Dashboard — Overview

The admin dashboard displays key business metrics including total revenue, number of orders, total customers, and total products. Real-time statistics are pulled dynamically from the database.

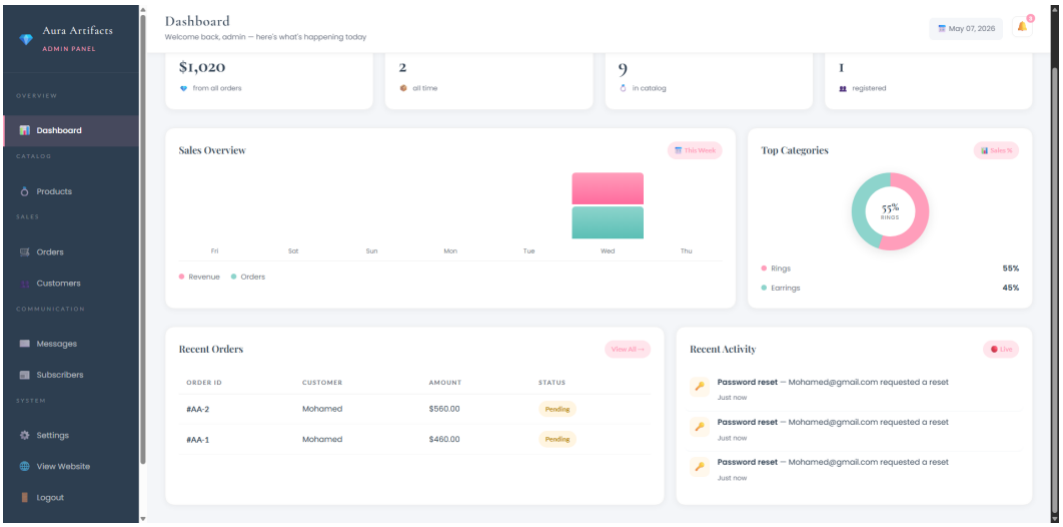


Figure 55: Aura Artifacts Admin Dashboard — Sales Analytics

The dashboard features interactive charts showing sales overview for the last 7 days and top product categories with visual percentage bars and a donut chart.

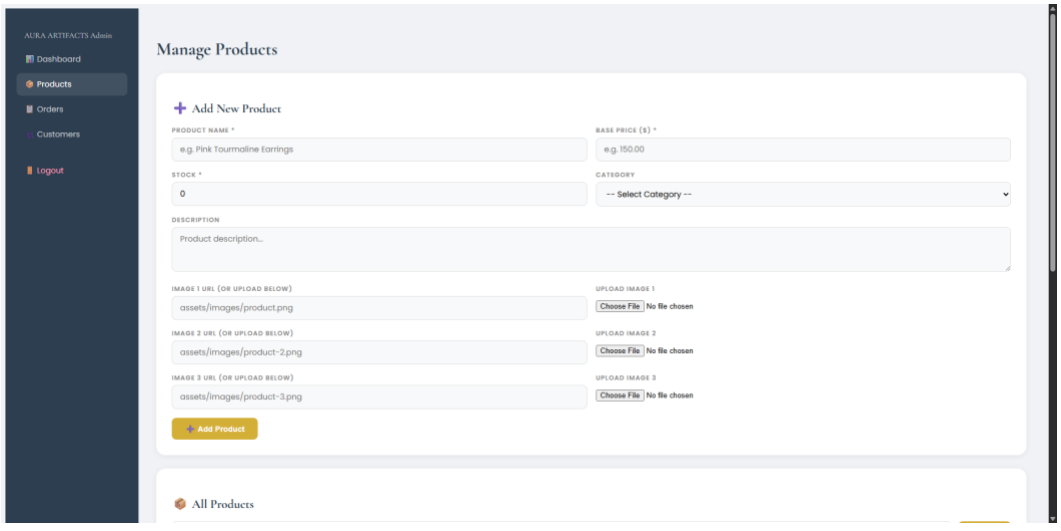


Figure 56: Aura Artifacts Admin — Products Management

The products management page allows administrators to view, add, edit, and delete products. The table displays product ID, image, name, category, price, and stock with action buttons.

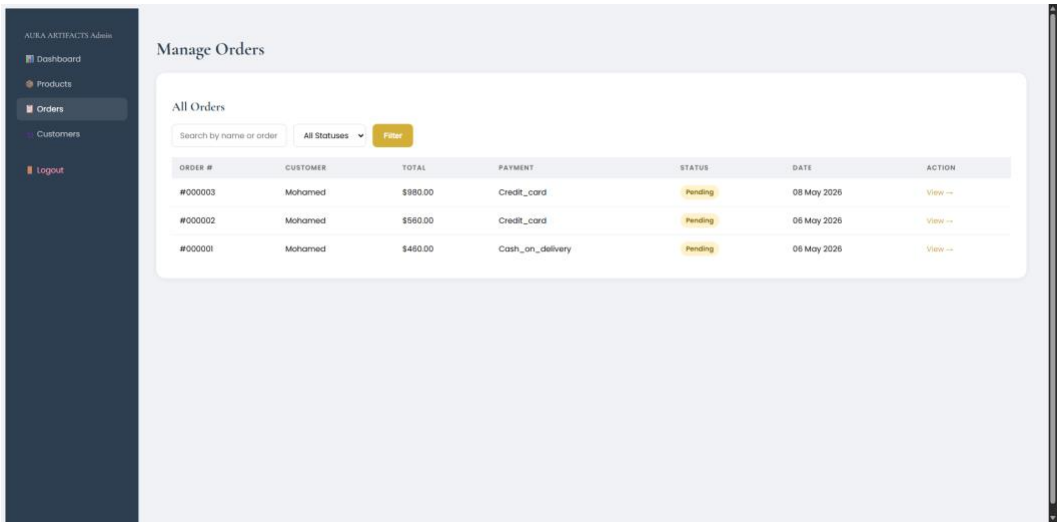


Figure 57: Aura Artifacts Admin — Orders Management

The orders management page shows all customer orders with order ID, customer name, date, total amount, status, and action options to update order status.

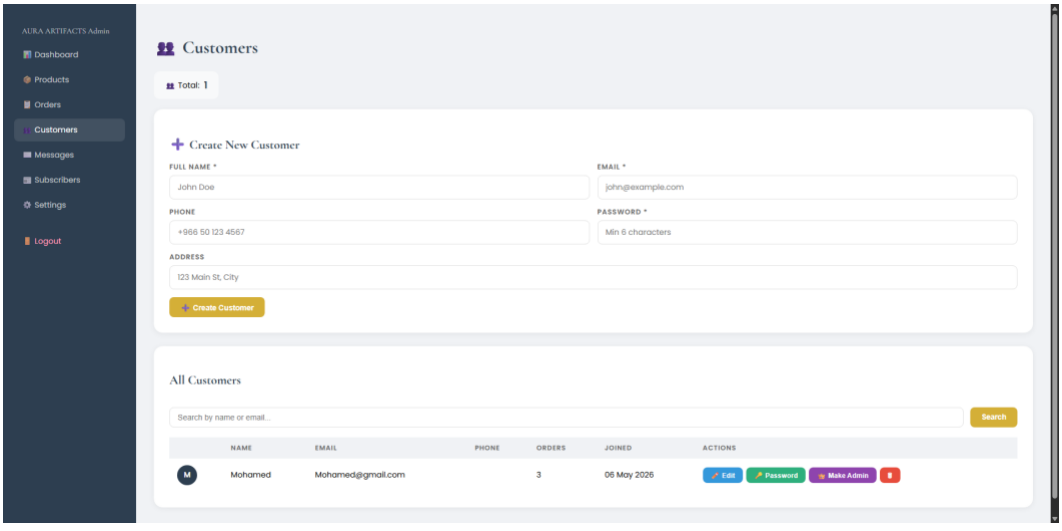


Figure 58: Aura Artifacts Admin — Customers Management

The customers page displays all registered users with their names, emails, phone numbers, registration dates, and total orders. Admins can create new customers, edit existing ones, change passwords, and delete accounts.

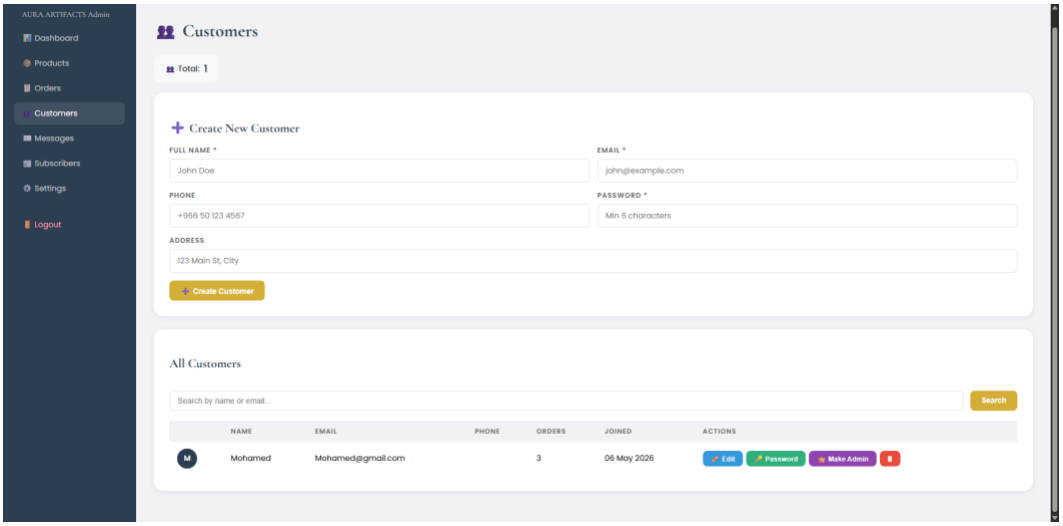


Figure 59: Aura Artifacts Admin — Make Admin / Make User Feature

The Make Admin feature allows administrators to promote regular users to admin role, granting them full access to the admin panel. The Make User button allows demoting admins back to regular user status, with a safety check preventing self-demotion.

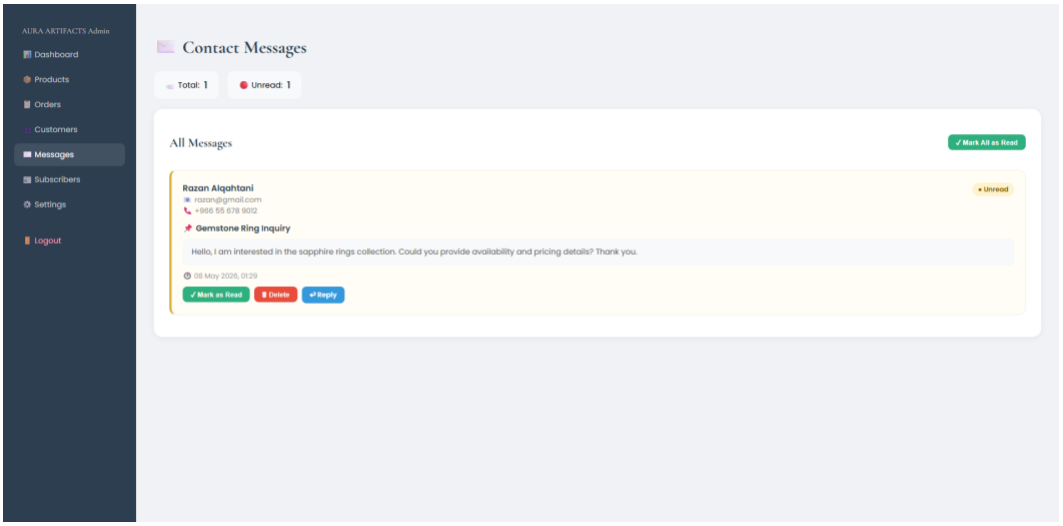


Figure 60: Aura Artifacts Admin — Contact Messages

The messages page shows all contact form submissions with sender name, email, subject, date, and read status. Admins can view full messages and mark them as read.

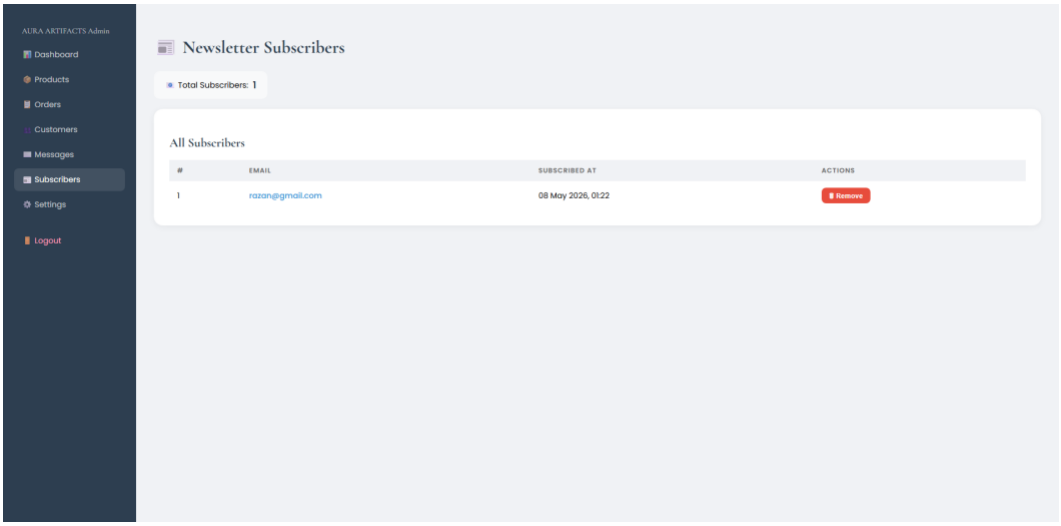


Figure 61: Aura Artifacts Admin — Newsletter Subscribers

The subscribers page manages the newsletter email list, showing subscriber emails and subscription dates.

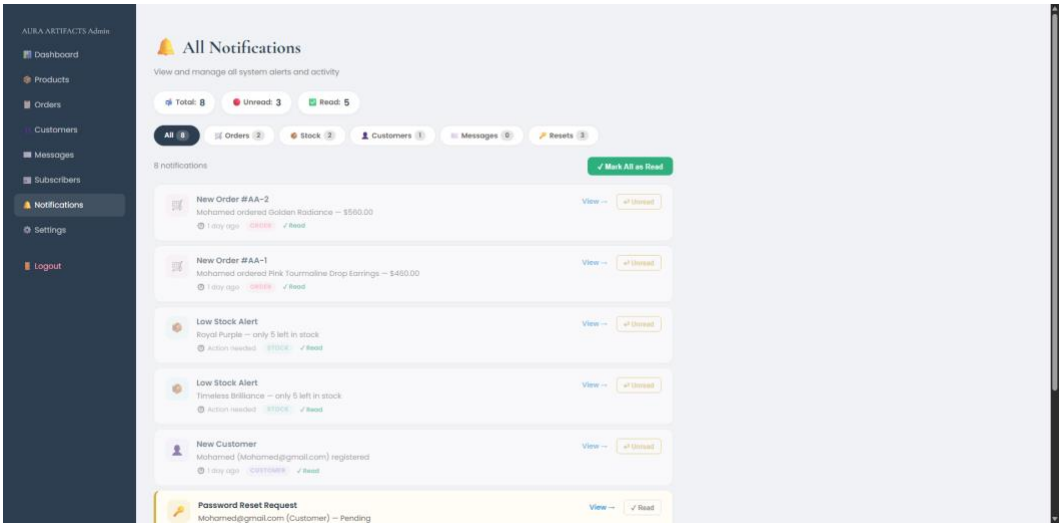


Figure 62: Aura Artifacts Admin — Notifications Center

The notifications center displays real-time alerts for new orders, customer registrations, contact messages, and low stock warnings.

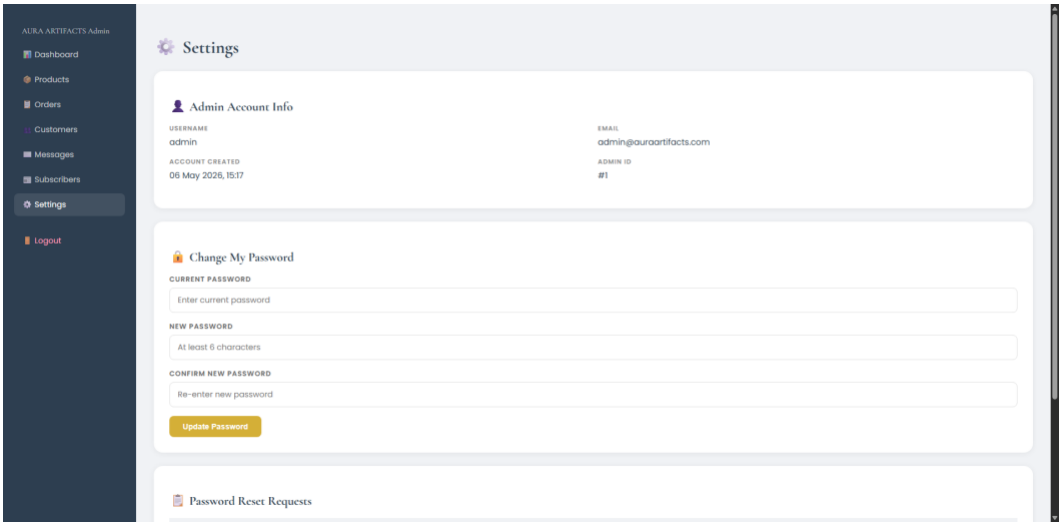


Figure 63: Aura Artifacts Admin — Settings

The settings page allows administrators to update their account information and manage system preferences.

3.4 Dark Mode Feature

Aura Artifacts includes a comprehensive dark mode feature that transforms the entire user interface to a dark color scheme. This enhances user comfort during nighttime browsing and reduces eye strain. The dark mode toggle is accessible from the navigation bar on every page.

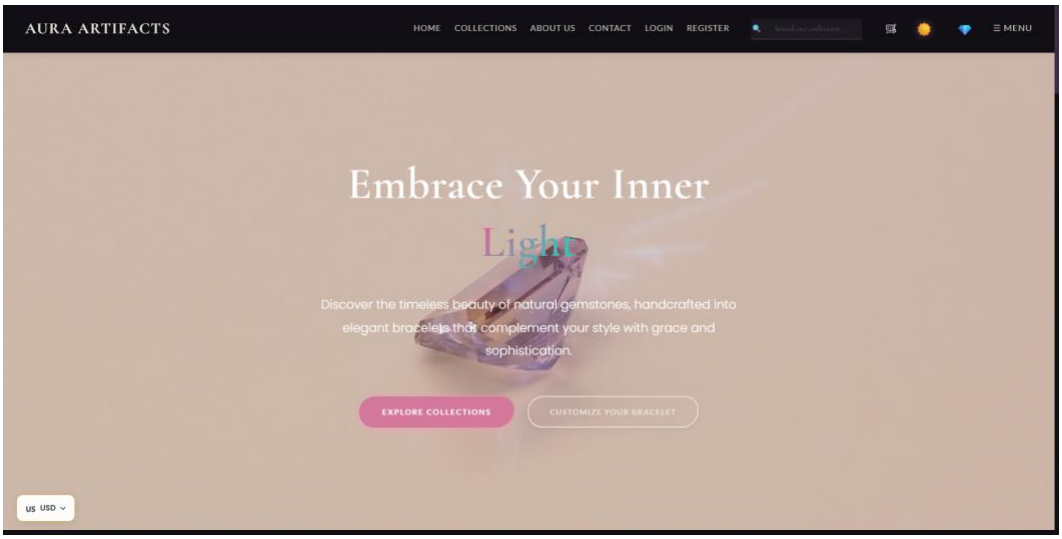


Figure 64: Aura Artifacts Homepage — Dark Mode

The homepage in dark mode with inverted colors, maintaining readability and visual hierarchy.

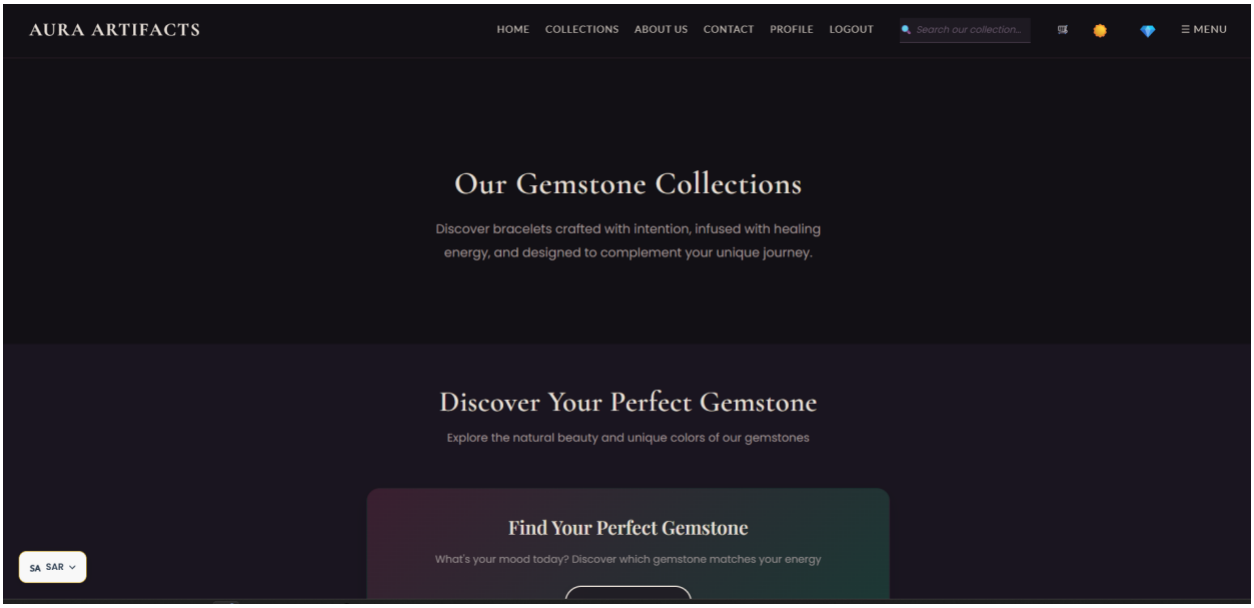


Figure 65: Aura Artifacts Collections Page — Dark Mode

The collections page in dark mode showing products with adapted styling for optimal contrast.

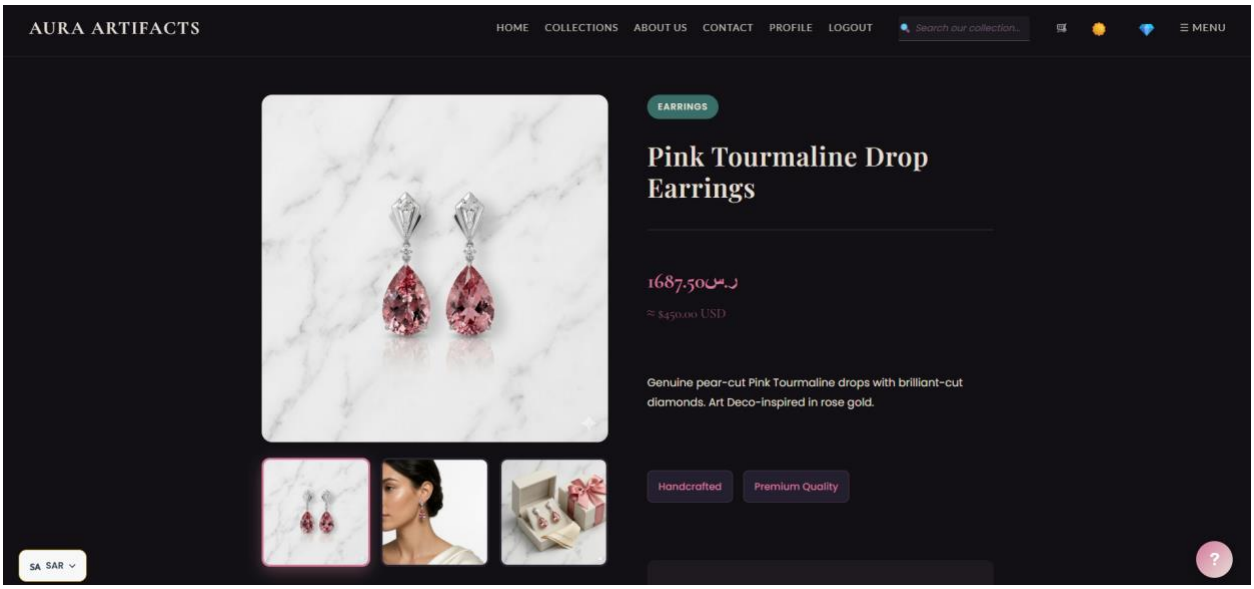


Figure 66: Aura Artifacts Product Detail — Dark Mode

The product detail page in dark mode demonstrating consistent dark theme across all elements.

3.5 Form Validation

The platform implements comprehensive form validation on both client-side and server-side to ensure data integrity and provide clear feedback to users.

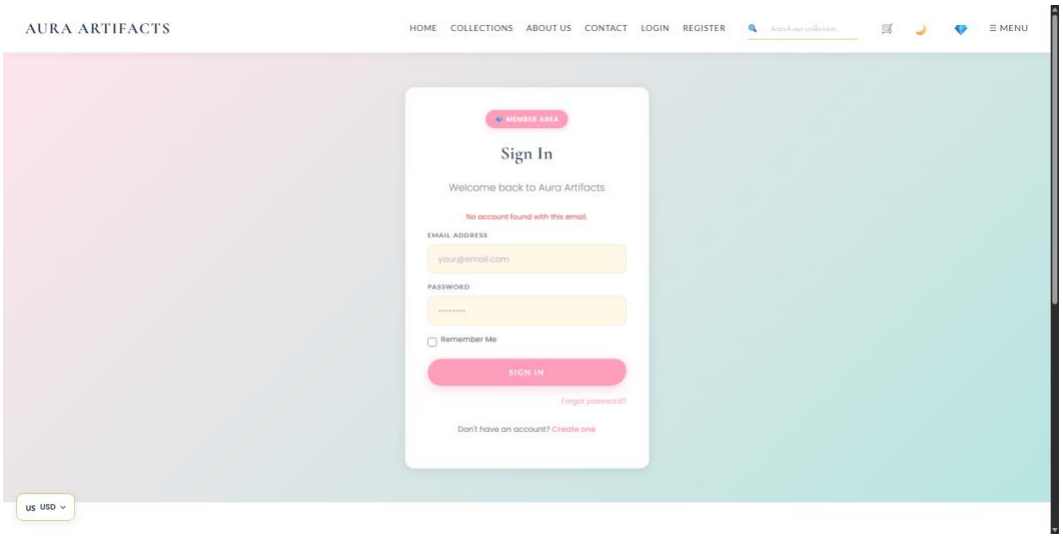


Figure 67: Aura Artifacts Login Page — Error Validation

The login page displays informative error messages when incorrect credentials are entered, helping users identify and correct their input.

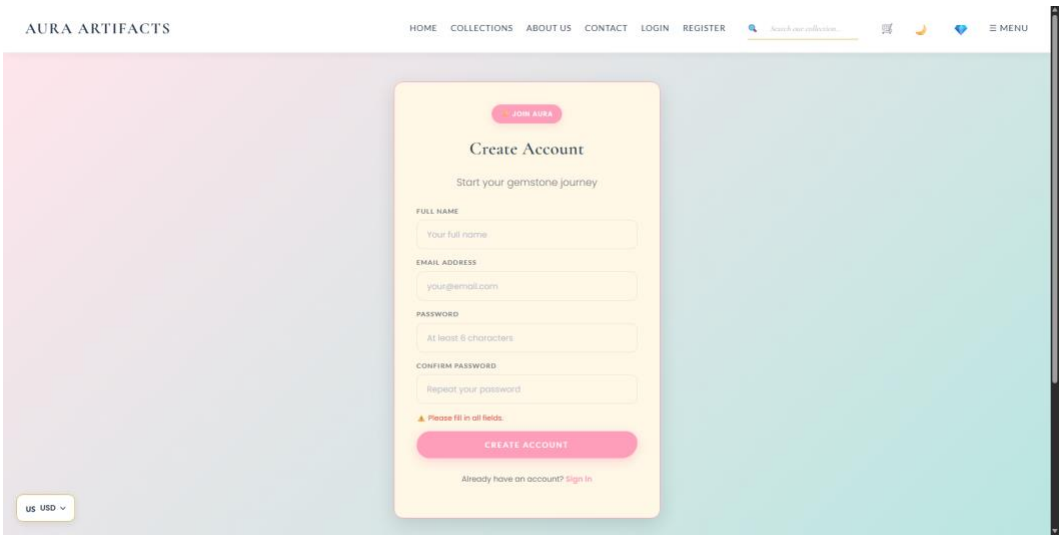


Figure 68: Aura Artifacts Register Page — Form Validation

The registration form validates all required fields including email format, password strength, and matching password confirmation before allowing submission.

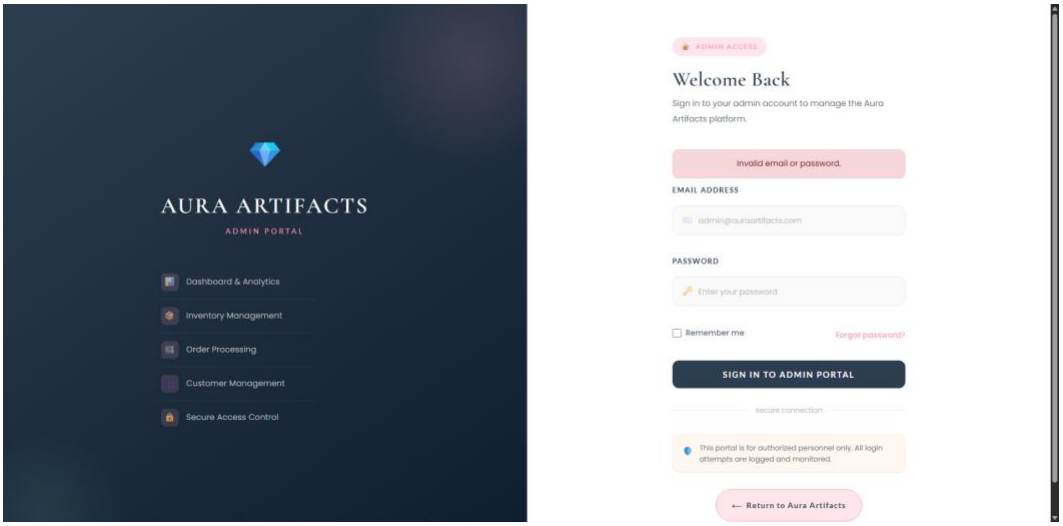


Figure 69: Aura Artifacts Admin Login — Error Validation

The admin login page shows validation errors for incorrect administrator credentials.

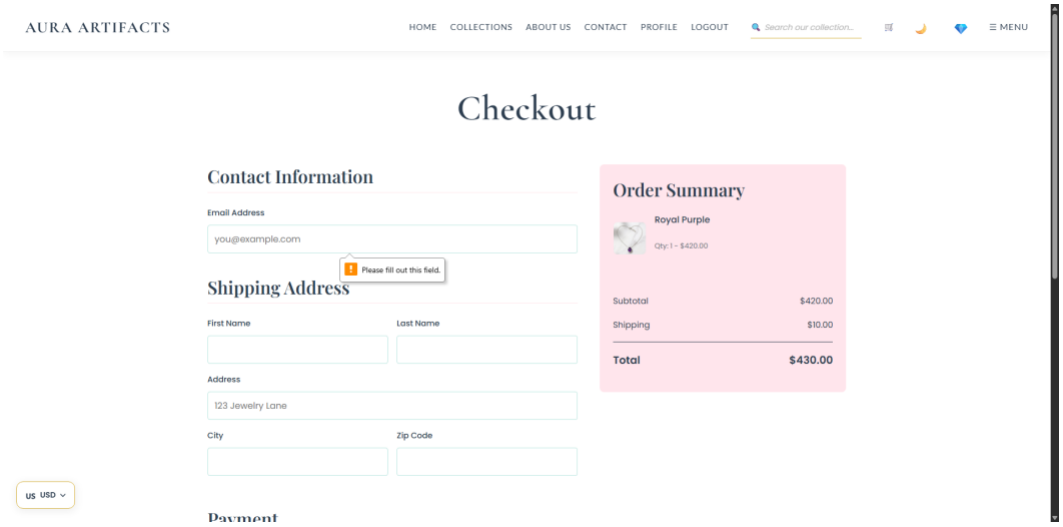


Figure 70: Aura Artifacts Checkout Page — Validation Errors

The checkout form validates all required shipping and payment fields before allowing order submission. Error messages are displayed inline next to each invalid field.

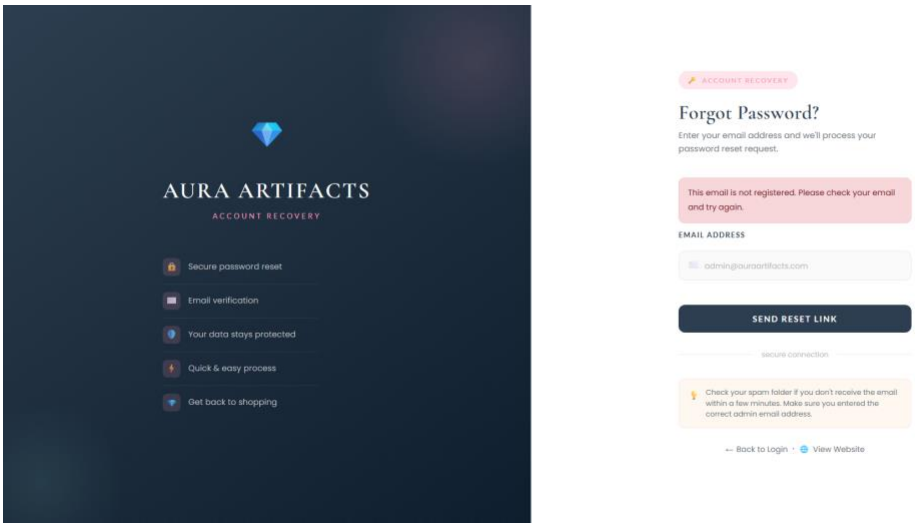


Figure 71: Aura Artifacts Forgot Password — Error Validation

The forgot password page shows a clear error message when a non-registered email address is entered.

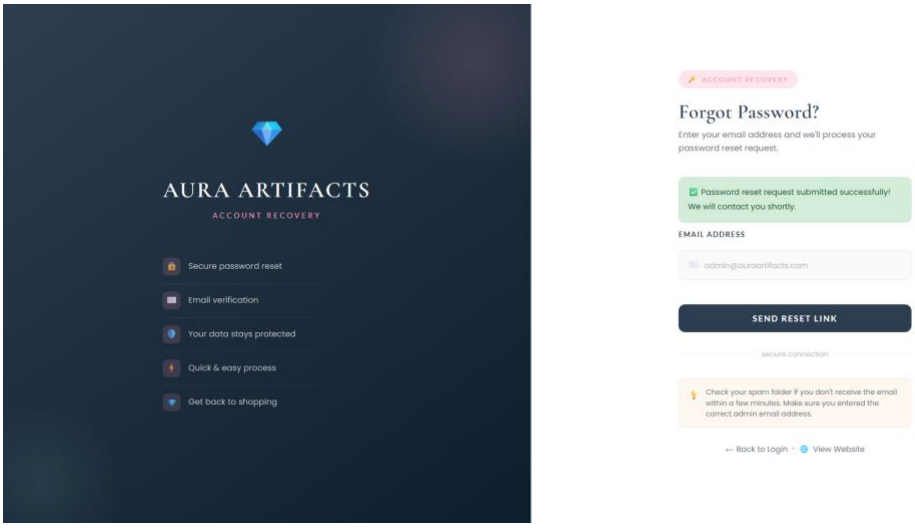


Figure 72: Aura Artifacts Forgot Password — Success

After entering a valid registered email, a success message confirms the password reset request has been submitted.

4. Fulfilled Requirements

Functional Requirements for the "Aura Artifacts" Platform

The "Aura Artifacts" platform is developed to deliver a comprehensive e-commerce experience for customers seeking handcrafted gemstone jewelry. The following sections outline the key functional requirements of the system.

1. User Management

• The system shall allow users to register using their email and a secure password.
• The system shall support secure login and logout mechanisms with session management.
• Users shall be able to update their profile information and change their passwords.
• The system shall support two account types: regular customer and system administrator.
• The system shall implement a 7-day 'Remember Me' feature using secure token-based cookies.

2. Product Catalog Management

• Users shall be able to browse available products with multiple images and detailed descriptions.
• Products shall be organized by categories (Bracelets, Earrings, Necklaces, Rings, Anklets, Pendants, Sunglasses).
• Users shall be able to search for products using the search bar.
• The system administrator shall be able to add, update, or delete products.

3. Shopping Cart & Checkout

• Users shall be able to add products to their shopping cart with quantity selection.
• The cart shall display real-time price calculations and allow quantity updates.
• The checkout process shall collect shipping address and credit card payment information.
• Users shall be able to view their complete order history with status tracking.

4. Wishlist Management

• Users shall be able to add and remove products from their wishlist.
• The wishlist shall be accessible from the navigation bar with a dropdown preview.

5. Custom Bracelet Builder

• Users shall be able to design custom bracelets by selecting gemstones and charms.
• The builder shall provide real-time preview and dynamic price calculation.

6. Contact & Communication

• The contact page shall include a form and Google Maps integration.
• Users shall be able to subscribe to the newsletter.
• Administrators shall be able to view and manage contact messages.

7. Admin Dashboard

• The dashboard shall display real-time sales analytics and business metrics.
• Administrators shall manage products, orders, customers, messages, and subscribers.
• The system shall provide notifications for new orders, registrations, and messages.

Non-Functional Requirements for the "Aura Artifacts" Platform

In addition to the functional capabilities, non-functional requirements define the quality attributes and operational constraints that ensure the system performs efficiently, securely, and reliably.

1. Performance Requirements:

• The platform shall support at least 30 concurrent users without noticeable performance degradation.
• All images shall be optimized for fast loading while maintaining visual quality.

2. Security Requirements:

• All user passwords shall be stored using secure encryption techniques (bcrypt).
• Administrative functions shall be protected with session-based authentication guards.
• All database queries shall use prepared statements to prevent SQL injection.
• CSRF protection shall be implemented on all forms.

3. Usability Requirements:

• The platform shall provide a user-friendly and intuitive interface with consistent layout.
• All error messages shall be informative and actionable, helping users correct their inputs.
• The website shall be fully responsive across desktop, tablet, and mobile devices.
• A dark mode option shall be available for user comfort.

4. Accessibility Requirements:

• The platform shall follow web accessibility best practices.
• All interactive elements shall have proper labels and focus states.
• The platform shall provide a dedicated accessibility information page.

5. Testing Cases

Test 1: Edit Customer

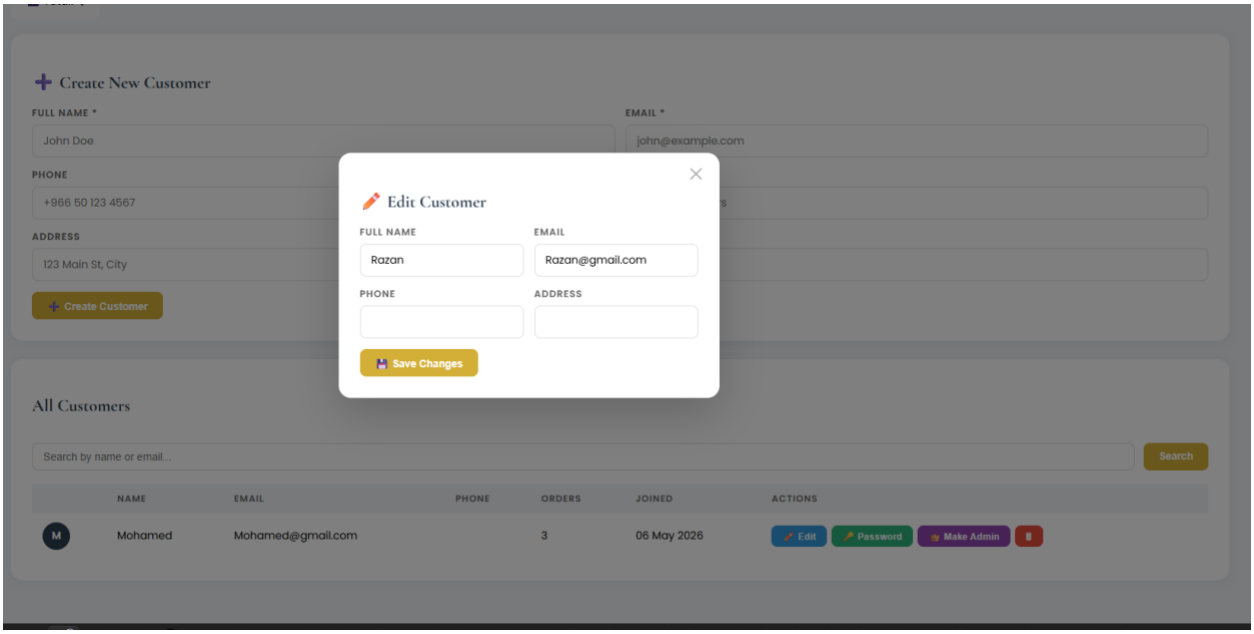


Figure 73: edit customer grid

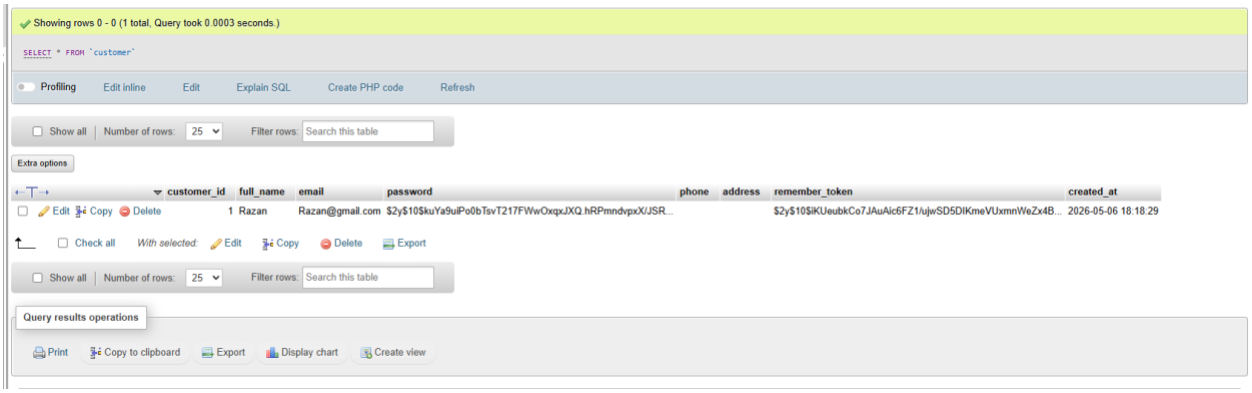


Figure 74: Result of editing customer in database

Test 2: Add Product

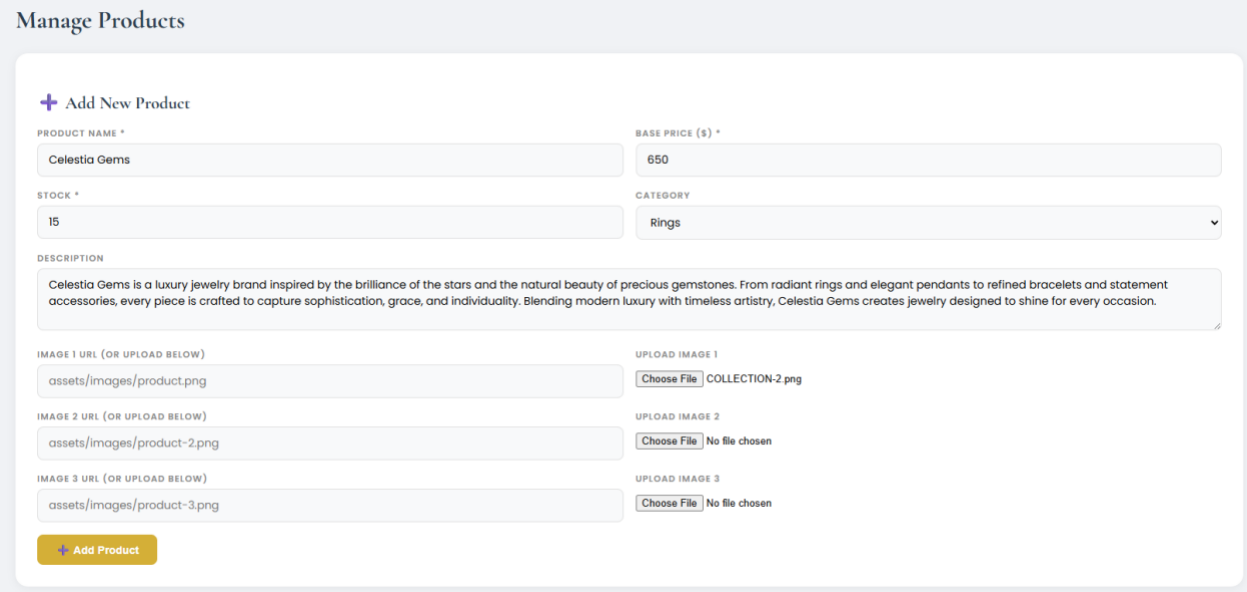
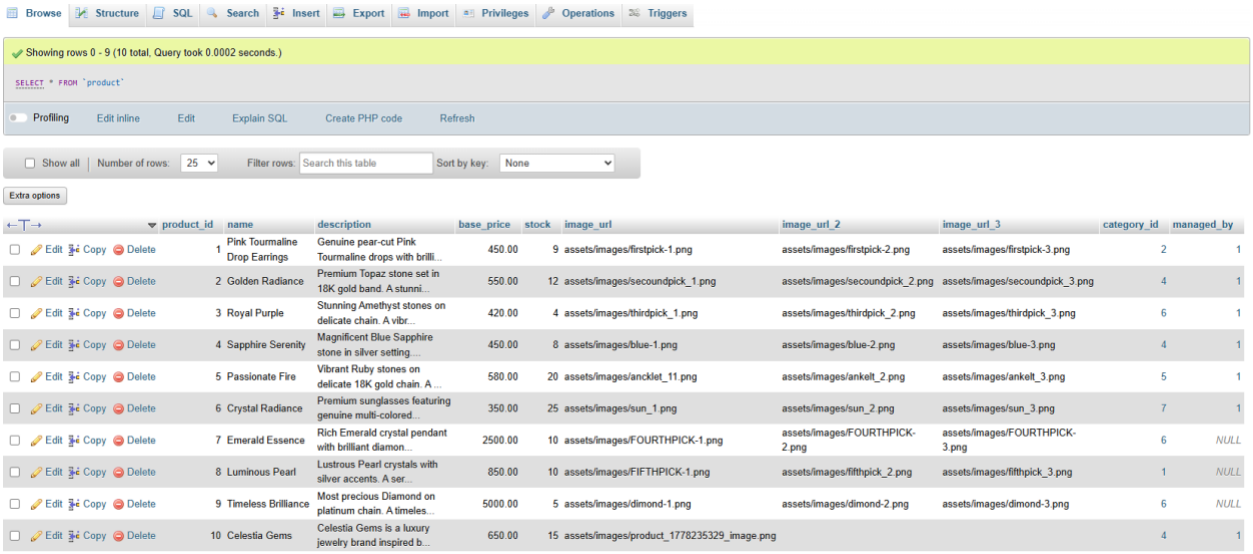


Figure75: adding a new product



	product_id	name	description	base_price	stock	image_url	image_url_2	image_url_3	category_id	managed_by
Edit Copy Delete	1	Pink Tourmaline Drop Earrings	Genuine pear-cut Pink Tourmaline drops with brilli...	450.00	9	assets/images/firstpick-1.png	assets/images/firstpick-2.png	assets/images/firstpick-3.png	2	1
Edit Copy Delete	2	Golden Radiance	Premium Topaz stone set in 18K gold band. A stunni...	550.00	12	assets/images/secoundpick_1.png	assets/images/secoundpick_2.png	assets/images/secoundpick_3.png	4	1
Edit Copy Delete	3	Royal Purple	Stunning Amethyst stones on delicate chain. A vibr...	420.00	4	assets/images/thirdpick_1.png	assets/images/thirdpick_2.png	assets/images/thirdpick_3.png	6	1
Edit Copy Delete	4	Sapphire Serenity	Magnificent Blue Sapphire stone in silver setting....	450.00	8	assets/images/blue-1.png	assets/images/blue-2.png	assets/images/blue-3.png	4	1
Edit Copy Delete	5	Passionate Fire	Vibrant Ruby stones on delicate 18K gold chain. A ...	580.00	20	assets/images/ancklet_11.png	assets/images/ankelt_2.png	assets/images/ankelt_3.png	5	1
Edit Copy Delete	6	Crystal Radiance	Premium sunglasses featuring genuine multi-colored...	350.00	25	assets/images/sun_1.png	assets/images/sun_2.png	assets/images/sun_3.png	7	1
Edit Copy Delete	7	Emerald Essence	Rich Emerald crystal pendant with brilliant diamon...	2500.00	10	assets/images/FOURTHPICK-1.png	assets/images/FOURTHPICK-2.png	assets/images/FOURTHPICK-3.png	6	NULL
Edit Copy Delete	8	Luminous Pearl	Lustrous Pearl crystals with silver accents. A ser...	850.00	10	assets/images/FIFTHPICK-1.png	assets/images/fifthpick_2.png	assets/images/fifthpick_3.png	1	NULL
Edit Copy Delete	9	Timeless Brilliance	Most precious Diamond on platinum chain. A timeles...	5000.00	5	assets/images/dimond-1.png	assets/images/dimond-2.png	assets/images/dimond-3.png	6	NULL
Edit Copy Delete	10	Celestia Gems	Celestia Gems is a luxury jewelry brand inspired b...	650.00	15	assets/images/product_1778235329_image.png			4	1

Figure 75: Result of adding a new product *in database*

Test 3: Manage Orders Status Update:

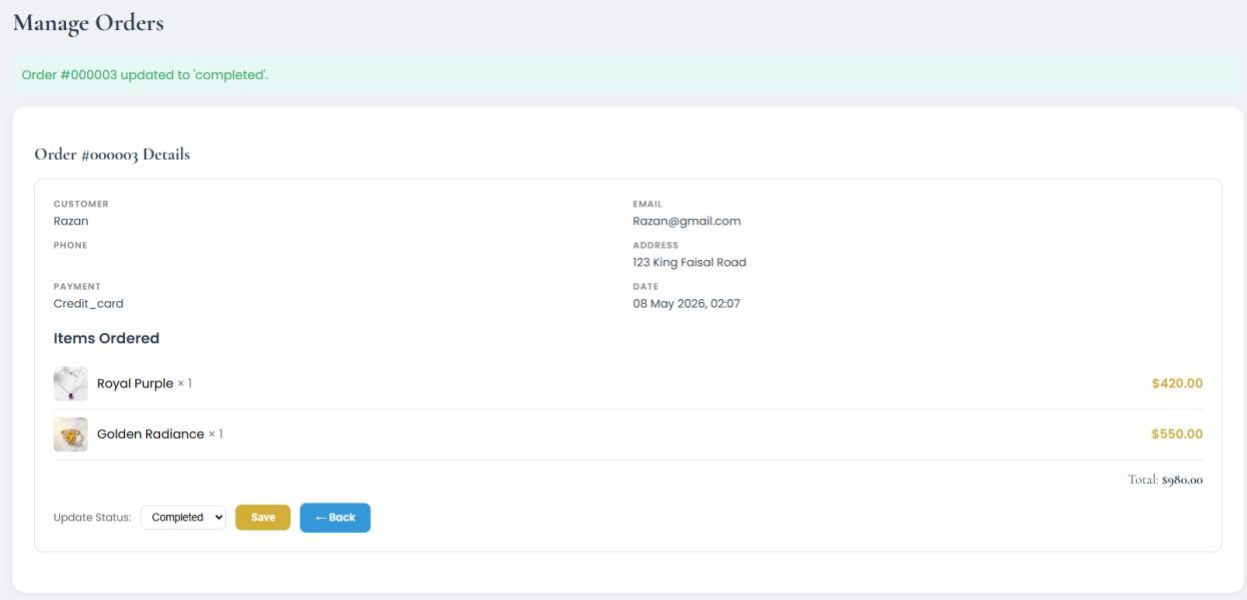


Figure76: editing an order status grid

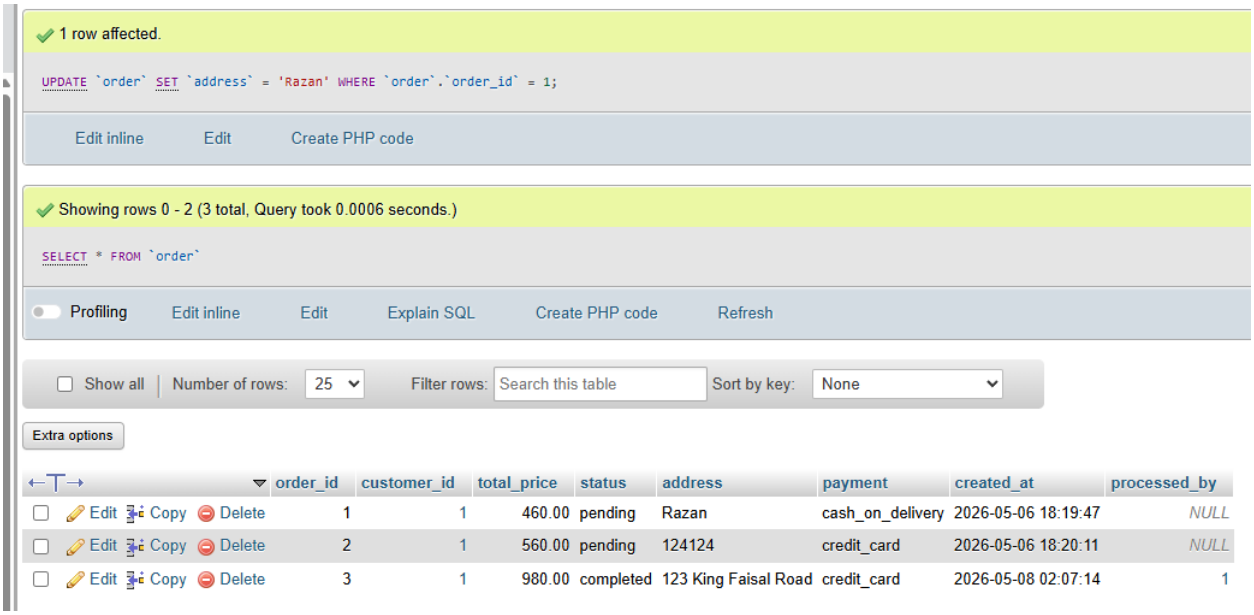


Figure 76: Result of editing an order status *in database*

Test 4: adding Customer

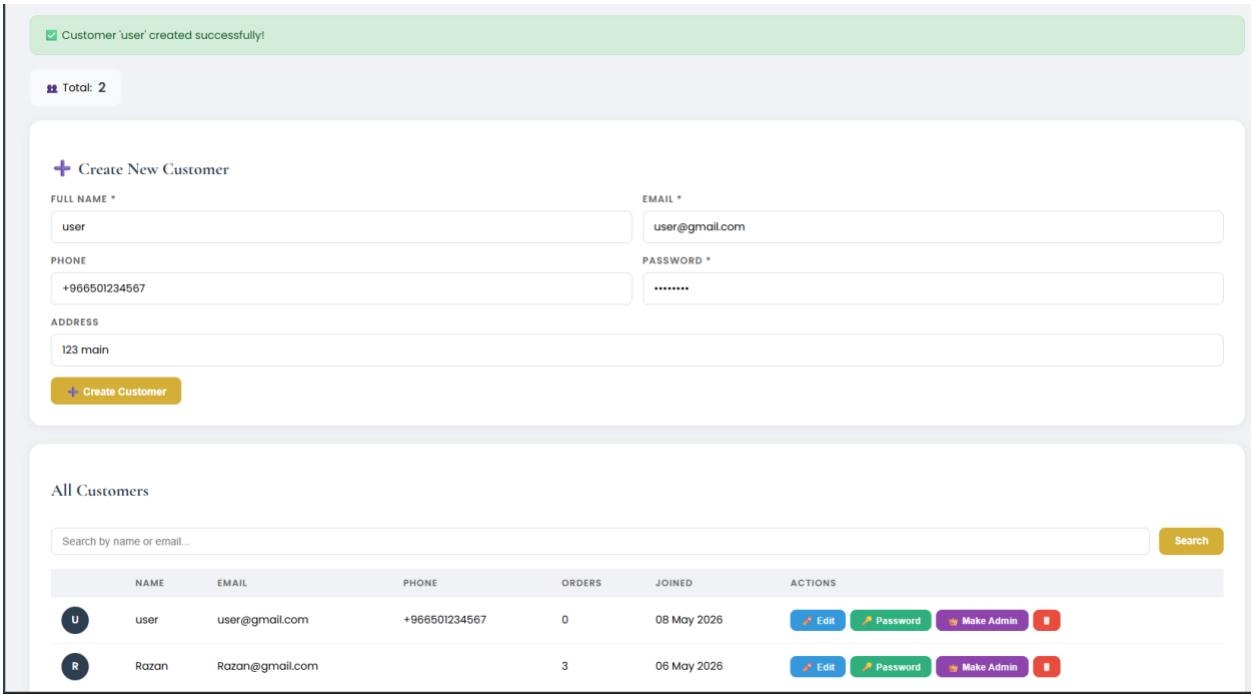


Figure77: adding a new customer Grid

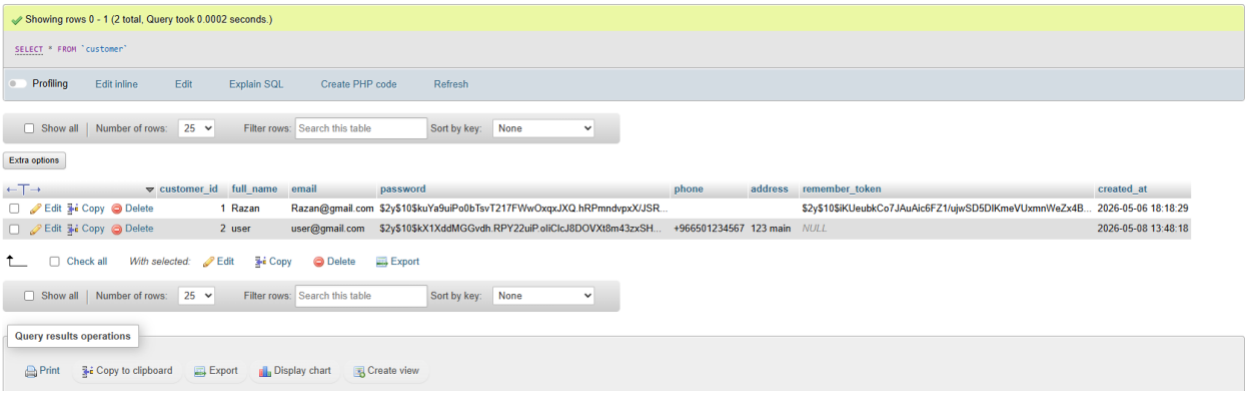


Figure 78: Result of adding a new customer in database

Test 5: Rest user password

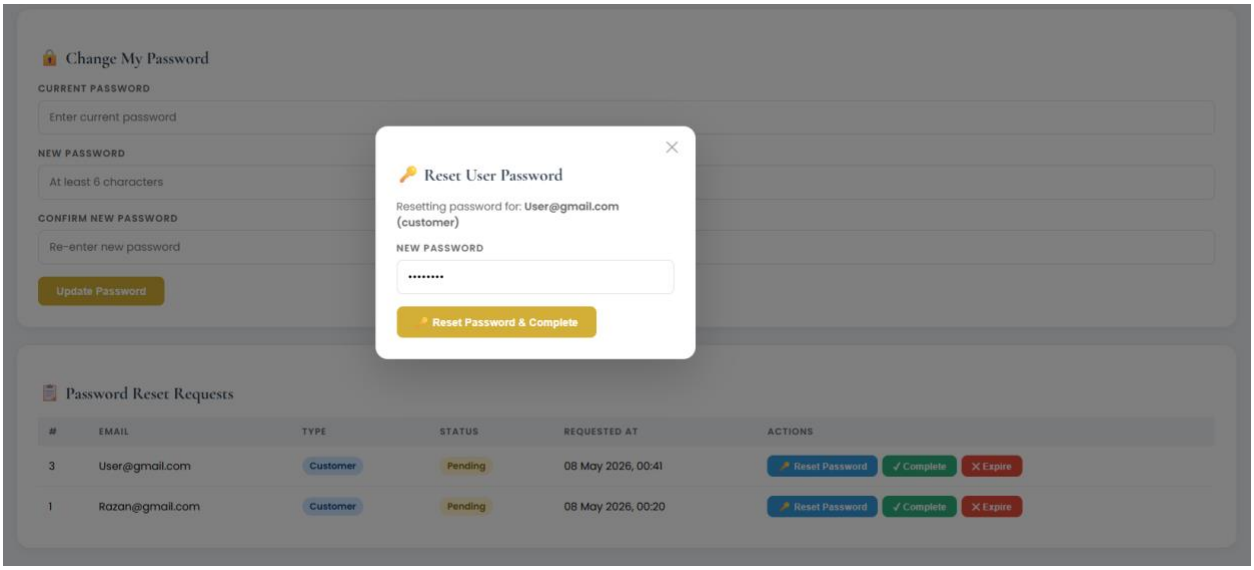


Figure 79: Rest user password grid

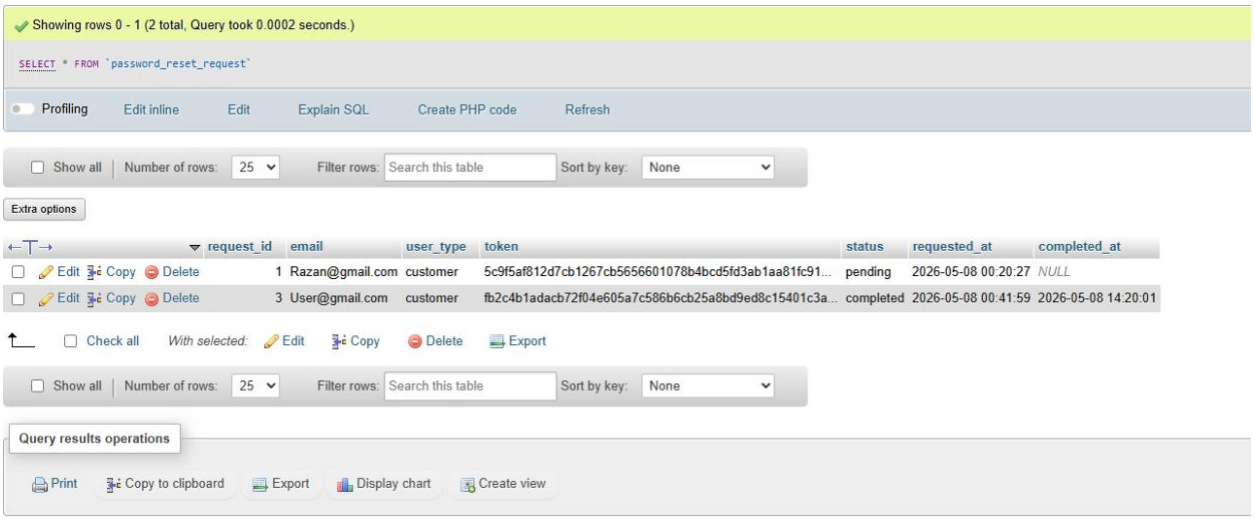


Figure 80: Result of rest user password in database

6. Team Collaboration

In web development, collaboration is crucial for building a successful project, and the Aura Artifacts platform demonstrates this effectively. The project was developed using a structured approach where the essential structure of the website was created using HTML5, which served as the foundation for all subsequent development.

The responsibilities were distributed among team members: front-end design and styling was handled using CSS3 with custom properties and responsive media queries, interactive features were implemented using vanilla JavaScript including dark mode, search functionality, and form validation, while the backend was built with PHP handling server-side logic, session management, and database operations through MySQLi with prepared statements. The database was designed and managed using MySQL with proper normalization and foreign key constraints.

Version control and consistent communication ensured the project moved forward seamlessly from initial planning through to the finished deployment-ready platform.

7. Conclusion

In conclusion, the Aura Artifacts platform was created to provide a premium online shopping experience for handcrafted gemstone jewelry enthusiasts. The primary goal of the platform is to establish a sophisticated e-commerce destination that combines beautiful design with robust functionality, offering features such as product browsing with multiple images, a custom bracelet

builder, secure checkout with credit card processing, user profiles with order history, and a comprehensive administrative dashboard.

The platform successfully implements all planned features including user authentication with remember-me functionality, a responsive design with dark mode support, real-time search, wishlist management, newsletter subscriptions, contact form with Google Maps integration, and an admin panel with sales analytics and complete store management capabilities.

Aura Artifacts demonstrates modern web development best practices including secure password hashing with bcrypt, SQL injection prevention through prepared statements, responsive design for all device sizes, semantic HTML for accessibility, and a clean separation of concerns between frontend presentation and backend logic. The platform is ready for production deployment and future expansion with additional product categories and features.